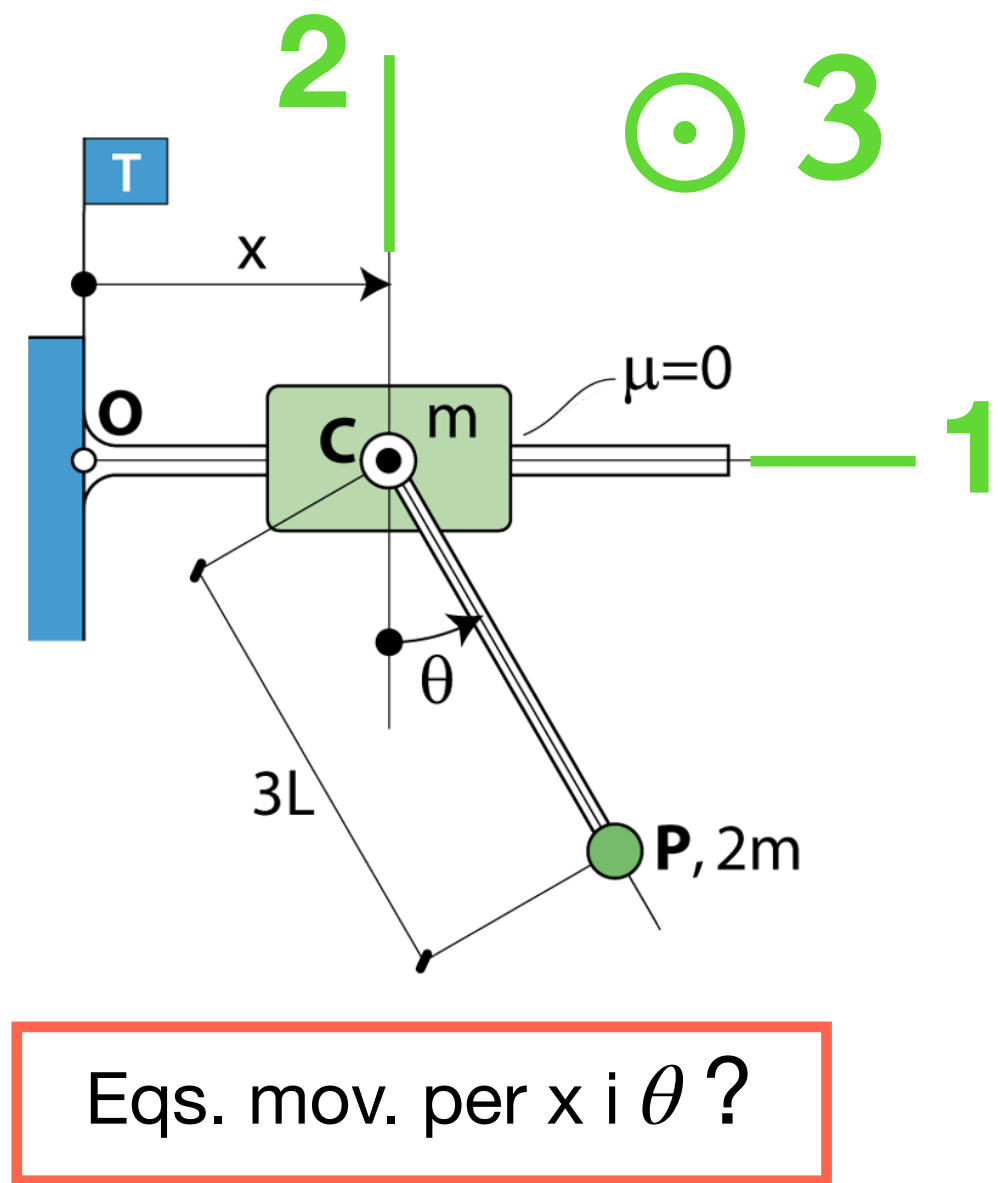
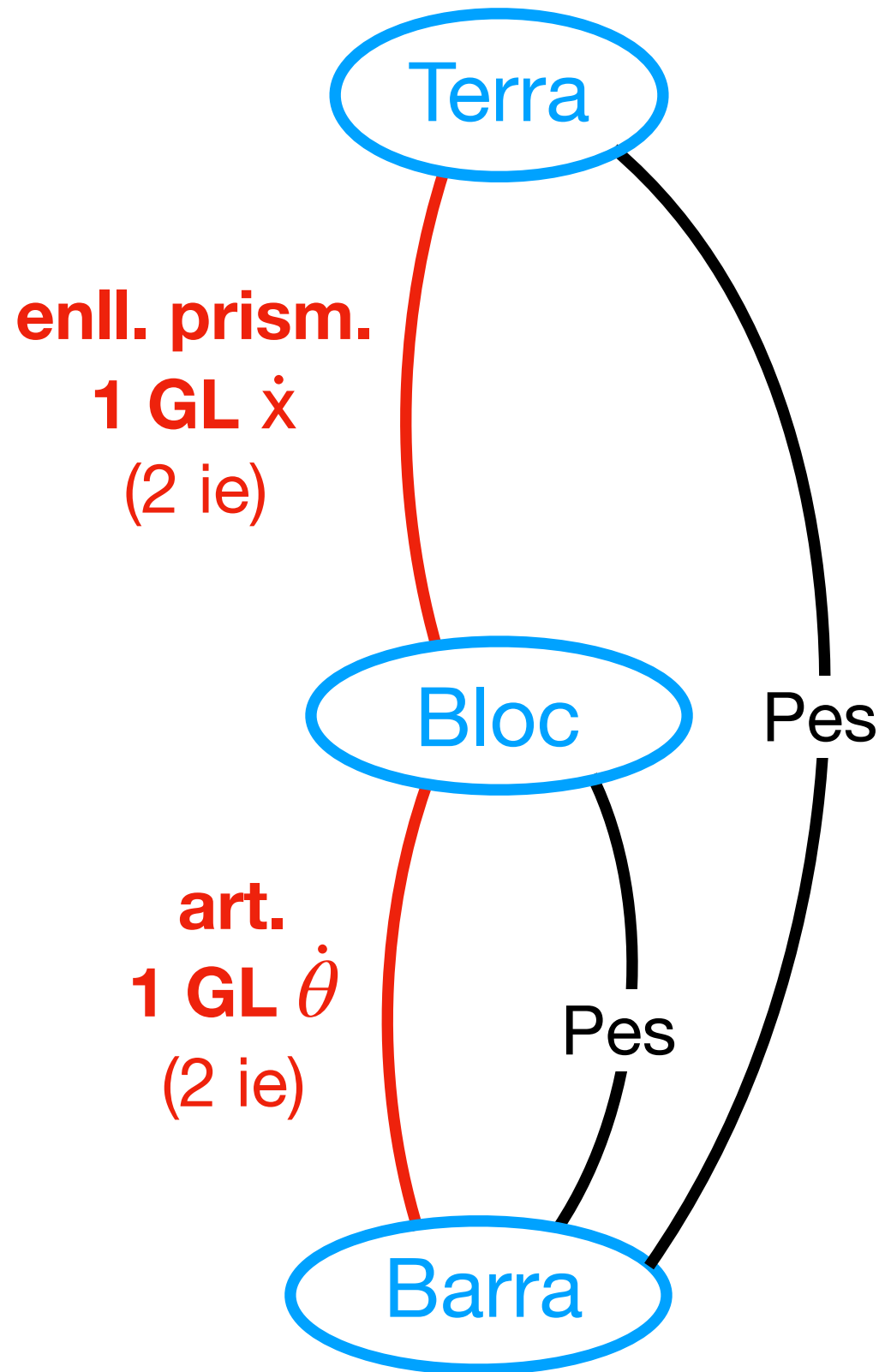


12P

Teoremes vectorials III

Exemples 3D

Pèndol d'Euler (recordem)



DGIE

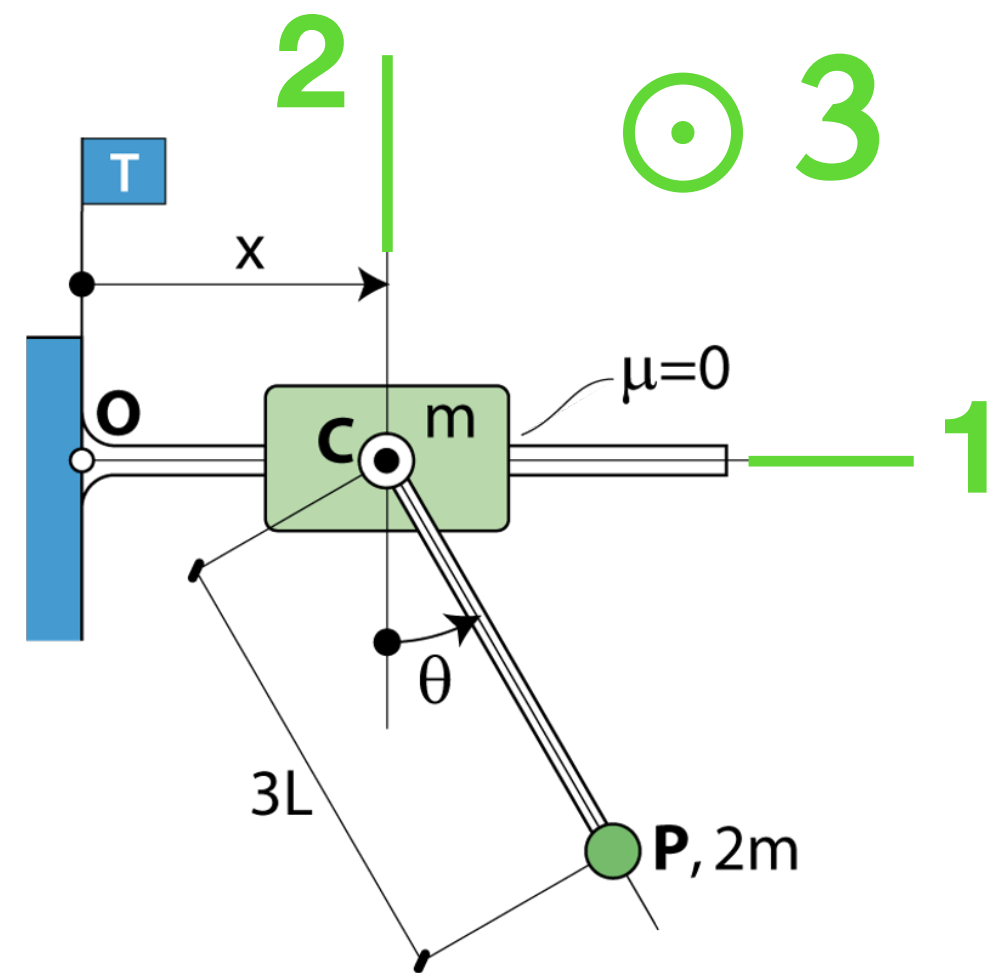
Terra

enll. prism.
1 GL $\dot{x}$
(2 ie)

Bloc

art.
1 GL $\dot{\theta}$
(2 ie)

Barra



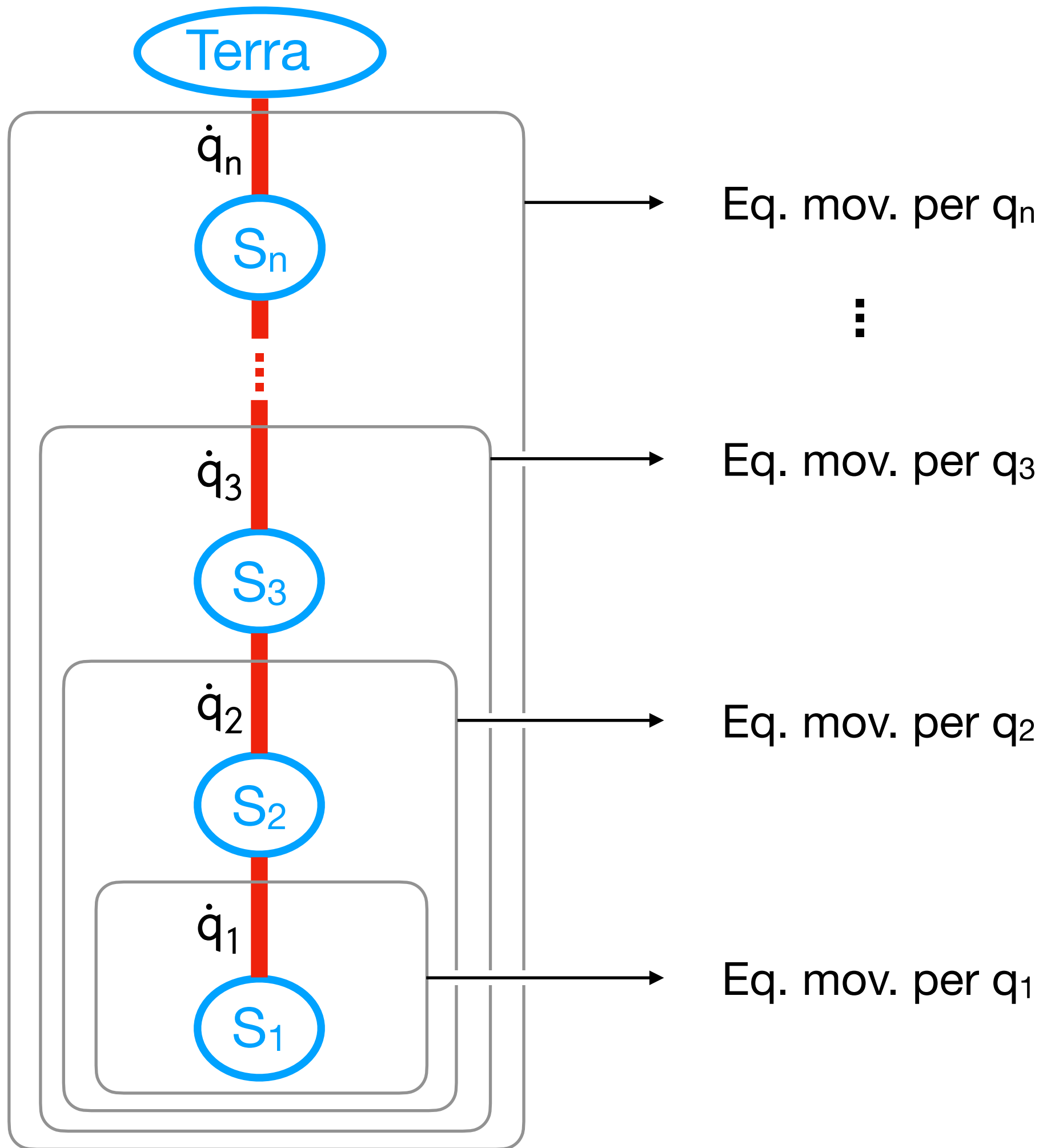
$$TQM]_1 = \text{dir. } \dot{x}$$

eq. mov.
per a x

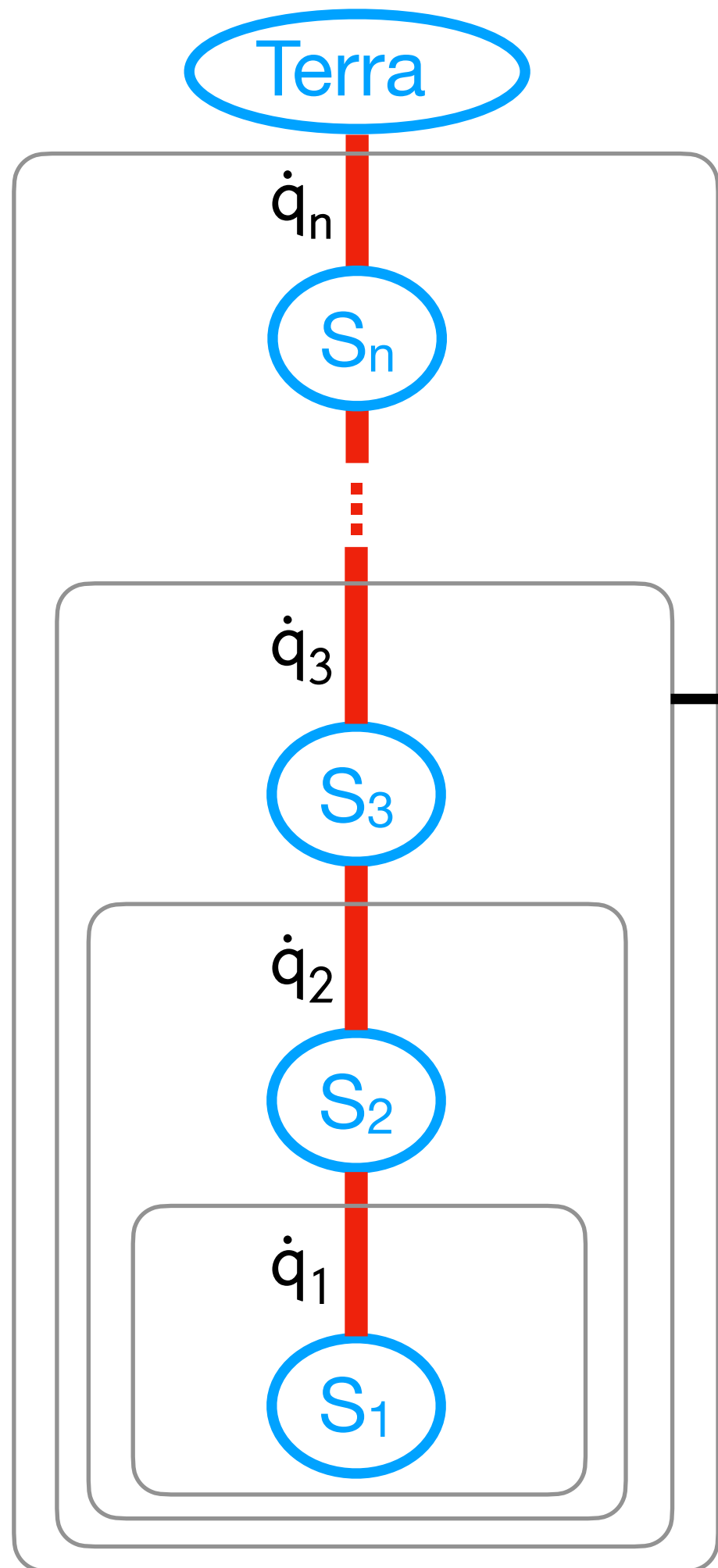
$$TMC(C)]_3 = \text{dir. } \dot{\theta}$$

eq. mov.
per a θ

En general



En general



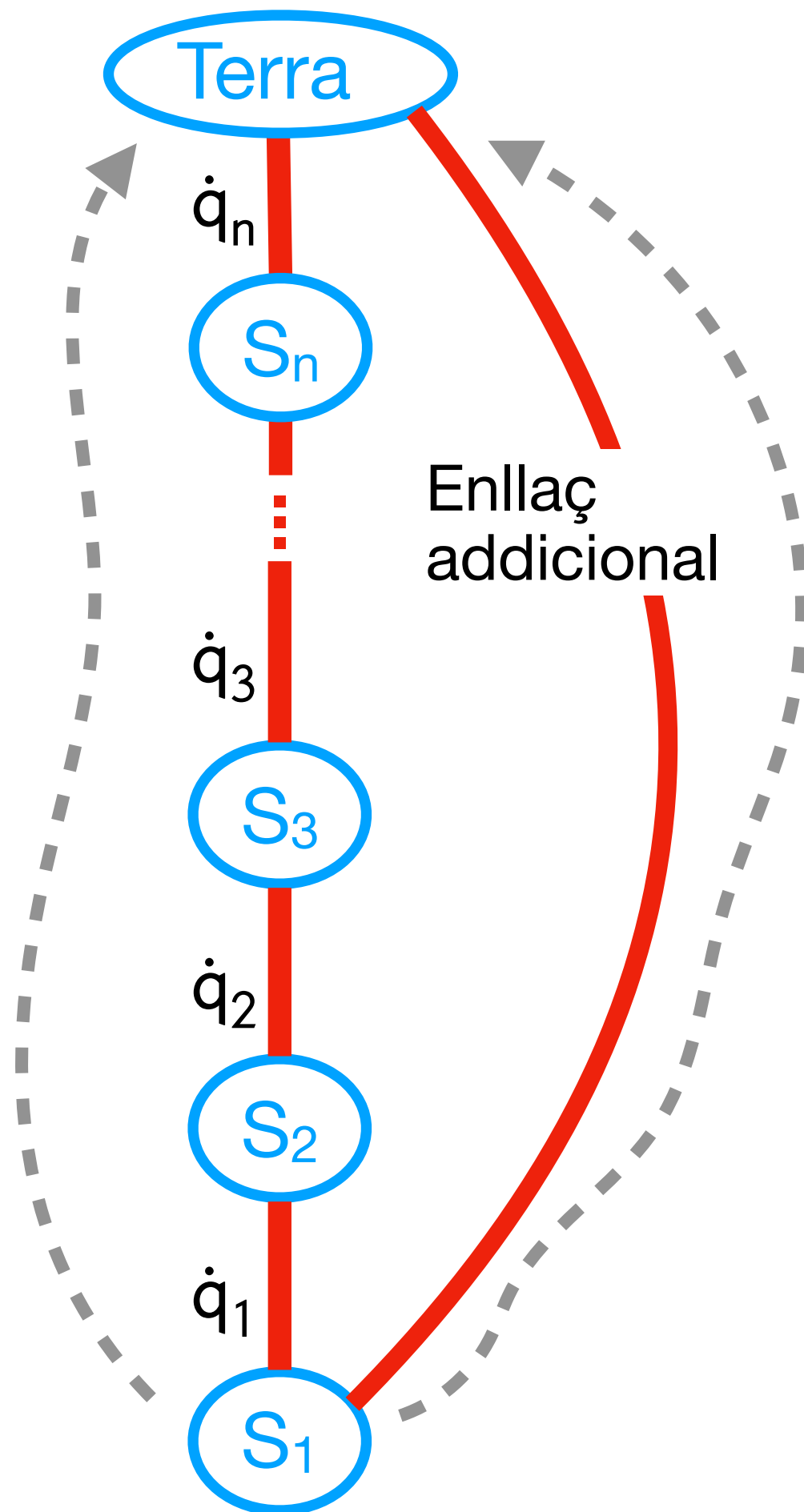
**I si algun GL
és forçat?**

Si un motor garanteix
 $\dot{q}_3 = ct$, el sistema

$$S_1 + S_2 + S_3$$

no dóna una eq. del mov,
sinó l'**acció motora**
(força o parell) que
assegura

$$\dot{q}_3 = ct$$



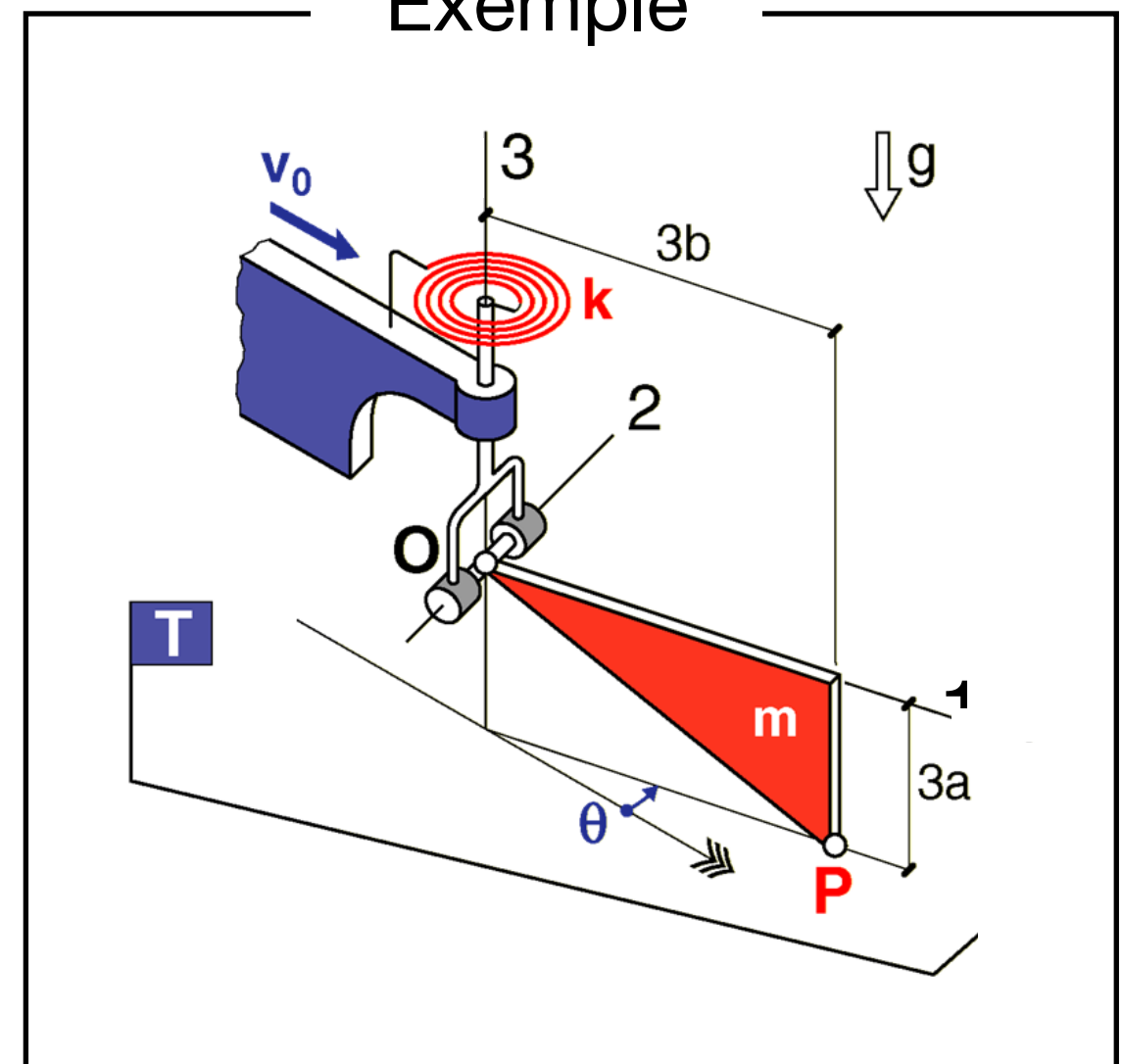
Compte!

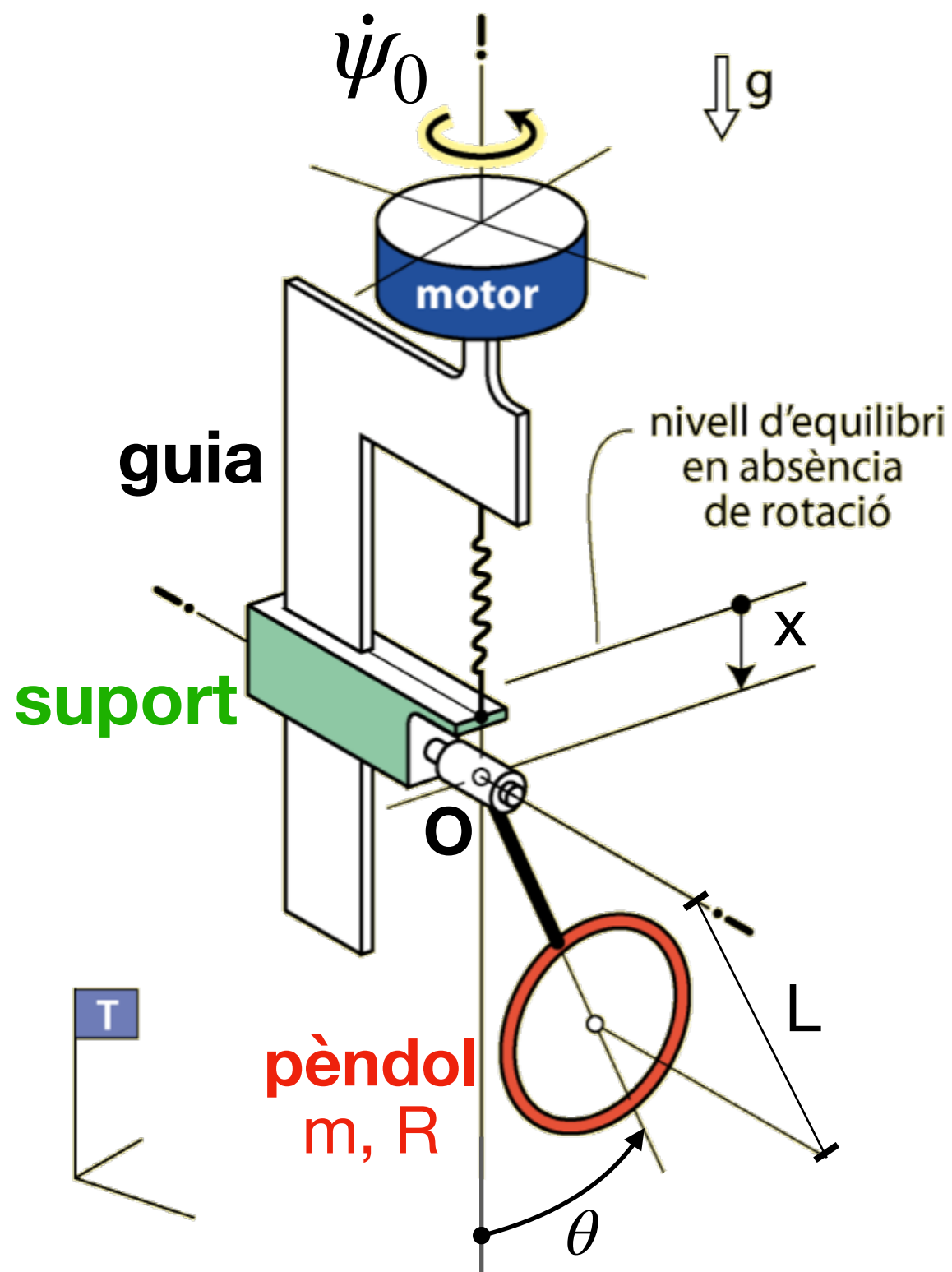
Camins múltiples entre S_1 i Terra



No podem aplicar els coneixements previs

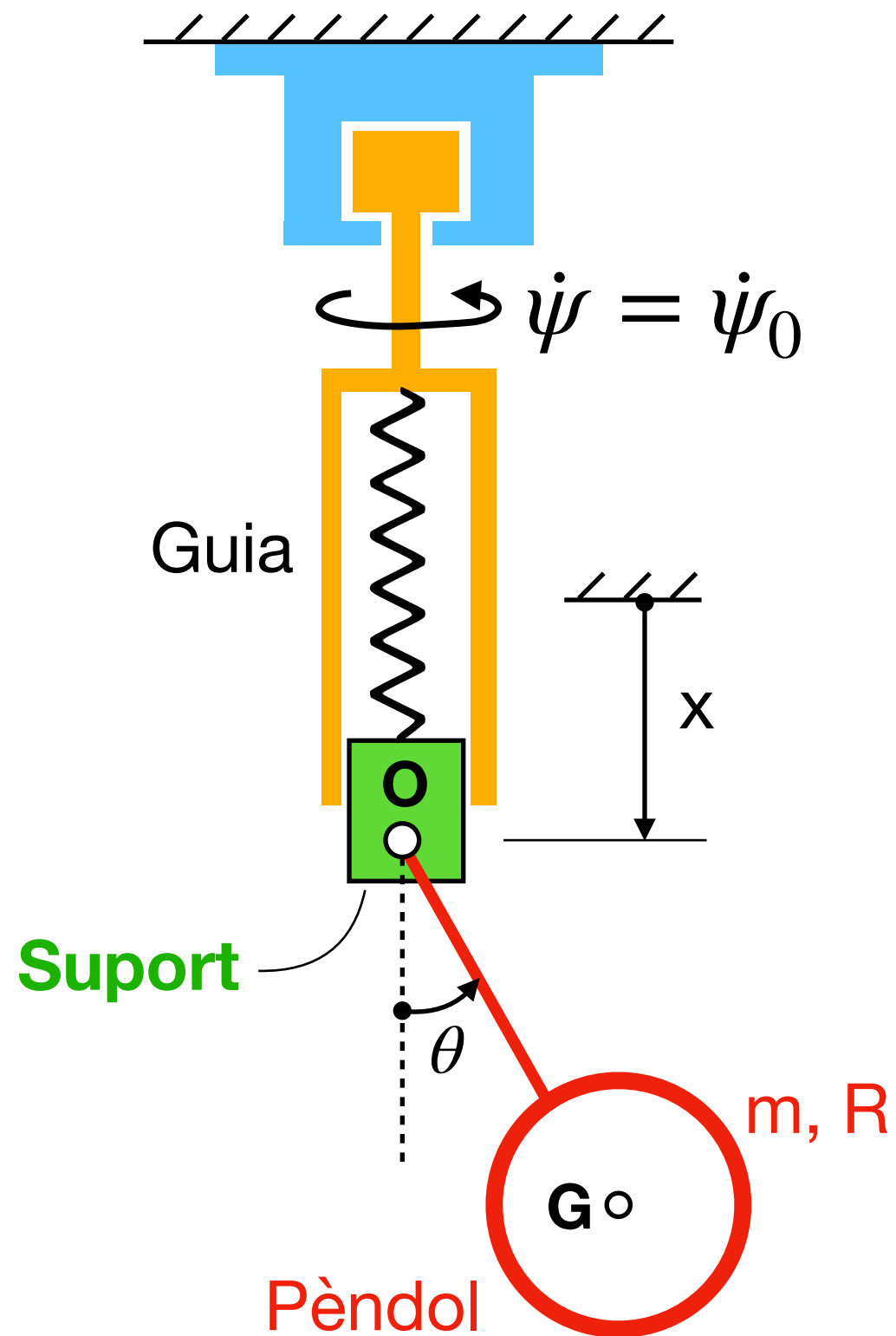
Exemple





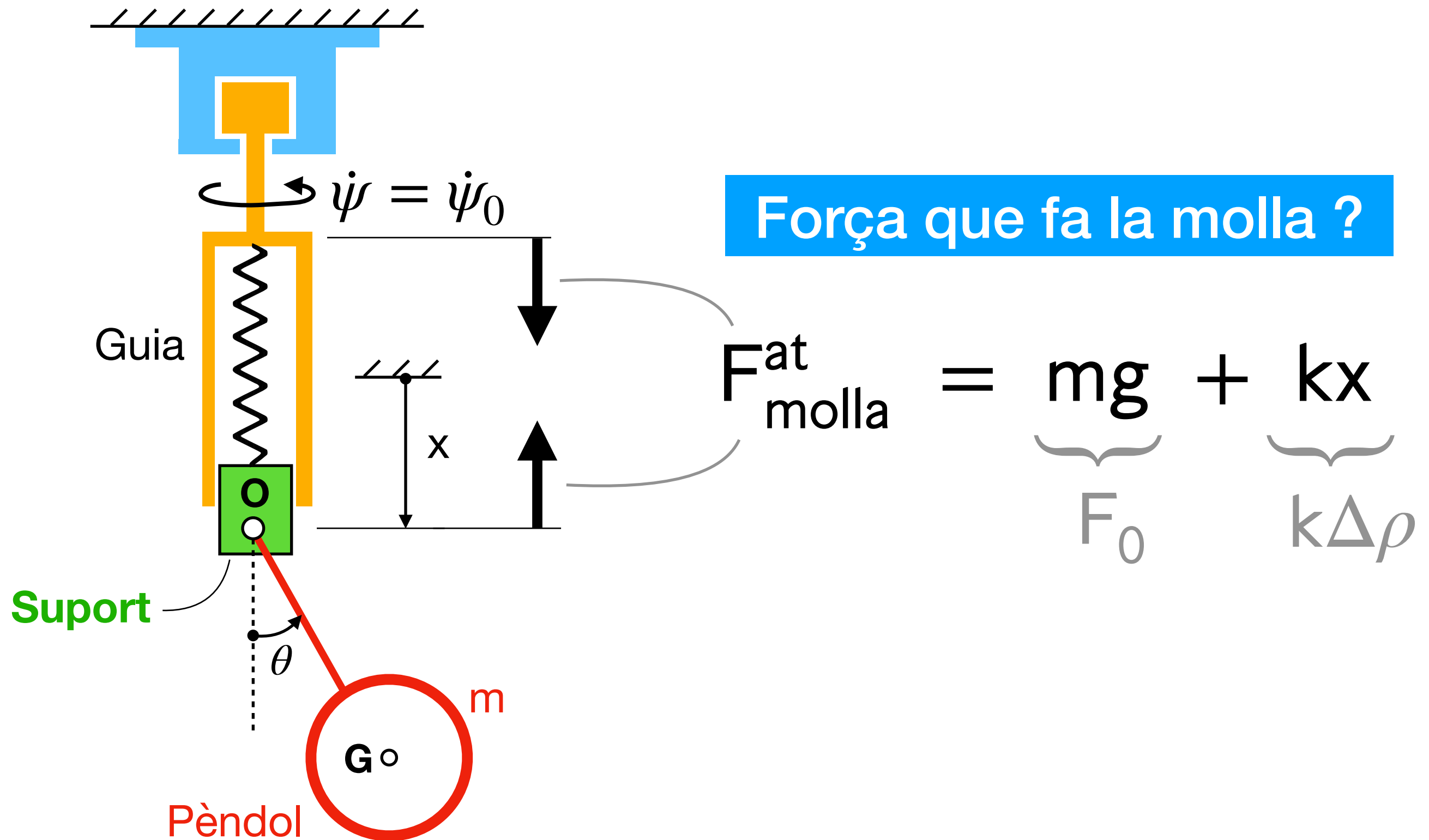
- DGI
- Eq. mov. per a x
- Eq. del mov per a θ
- Parell motor Γ per mantenir $\dot{\psi}_0 = ct$

Amb motor aturat: $x = 0$, $\theta = 0$ és config. d'equilibri



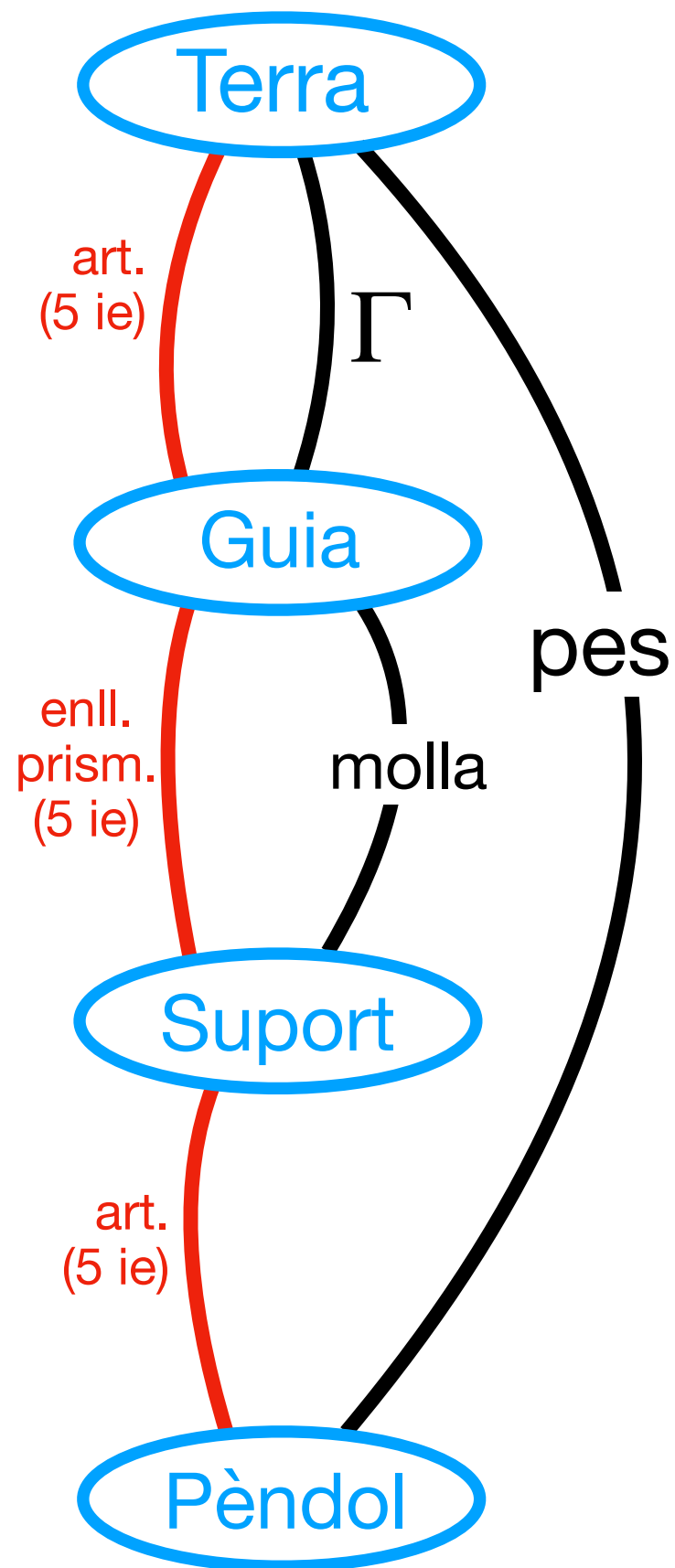
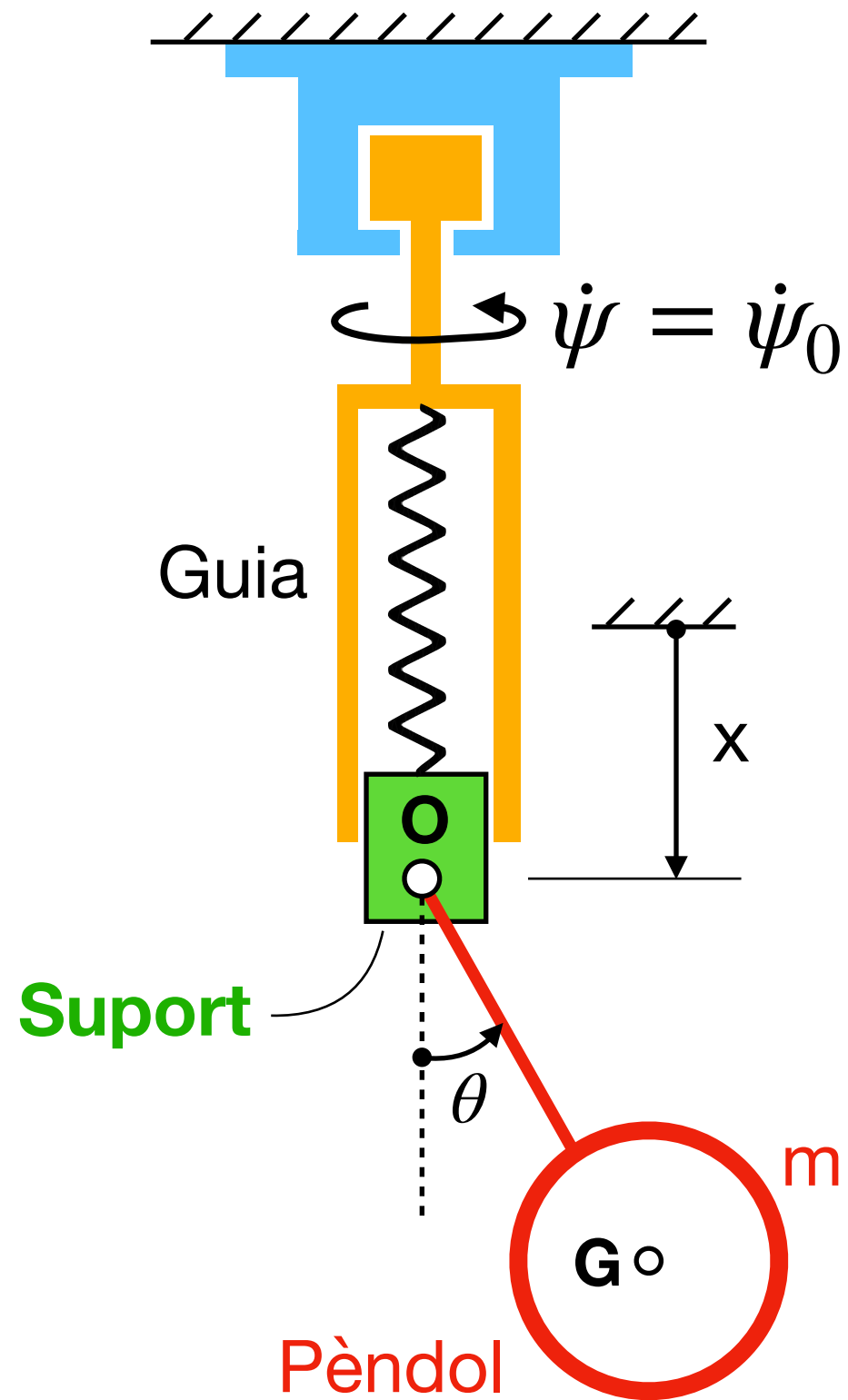
- DGI
- Eq. mov. per a x
- Eq. del mov per a θ
- Parell motor Γ per mantenir $\dot{\psi}_0 = ct$

Amb motor aturat: $x = 0$, $\theta = 0$ és config. d'equilibri

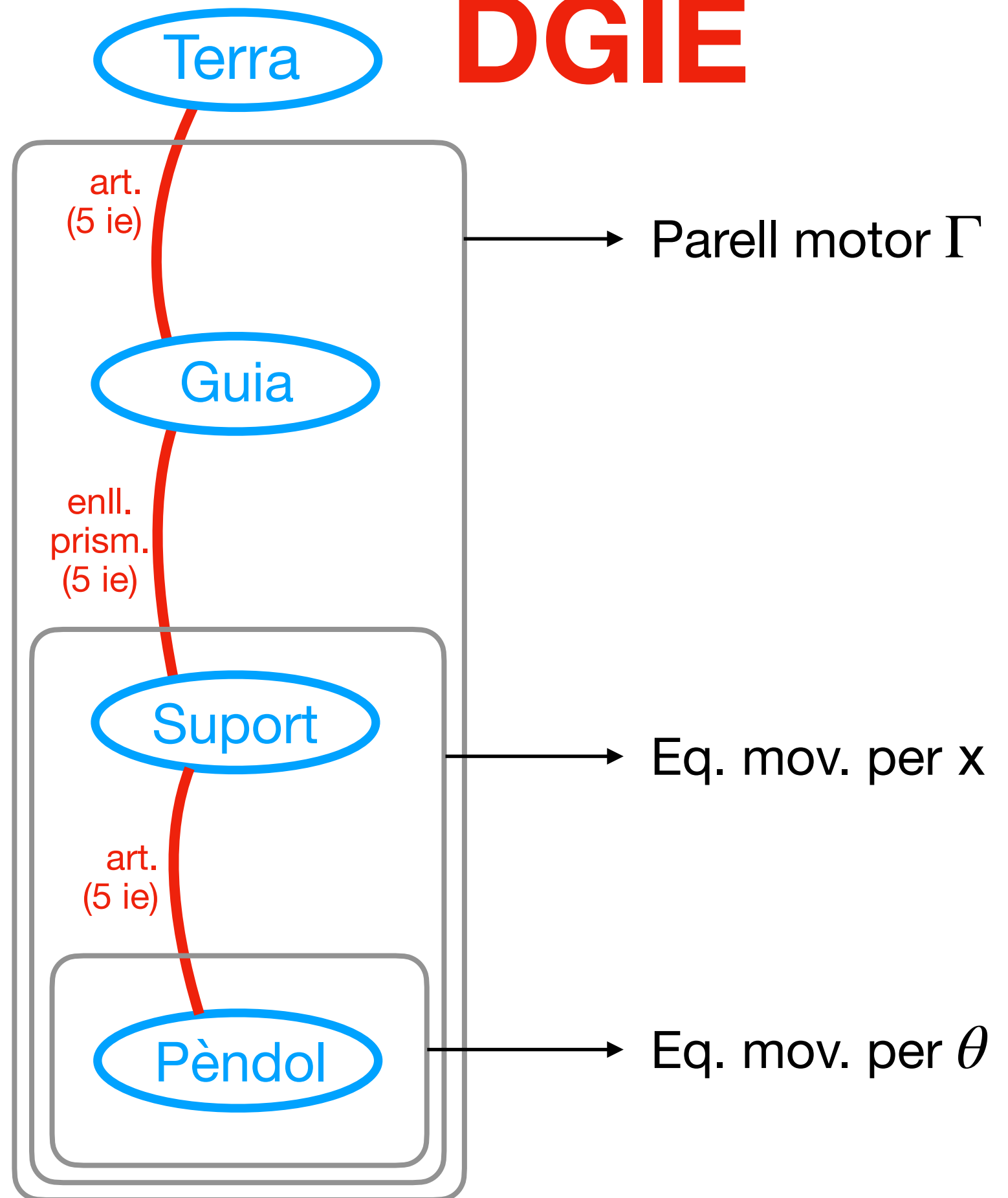
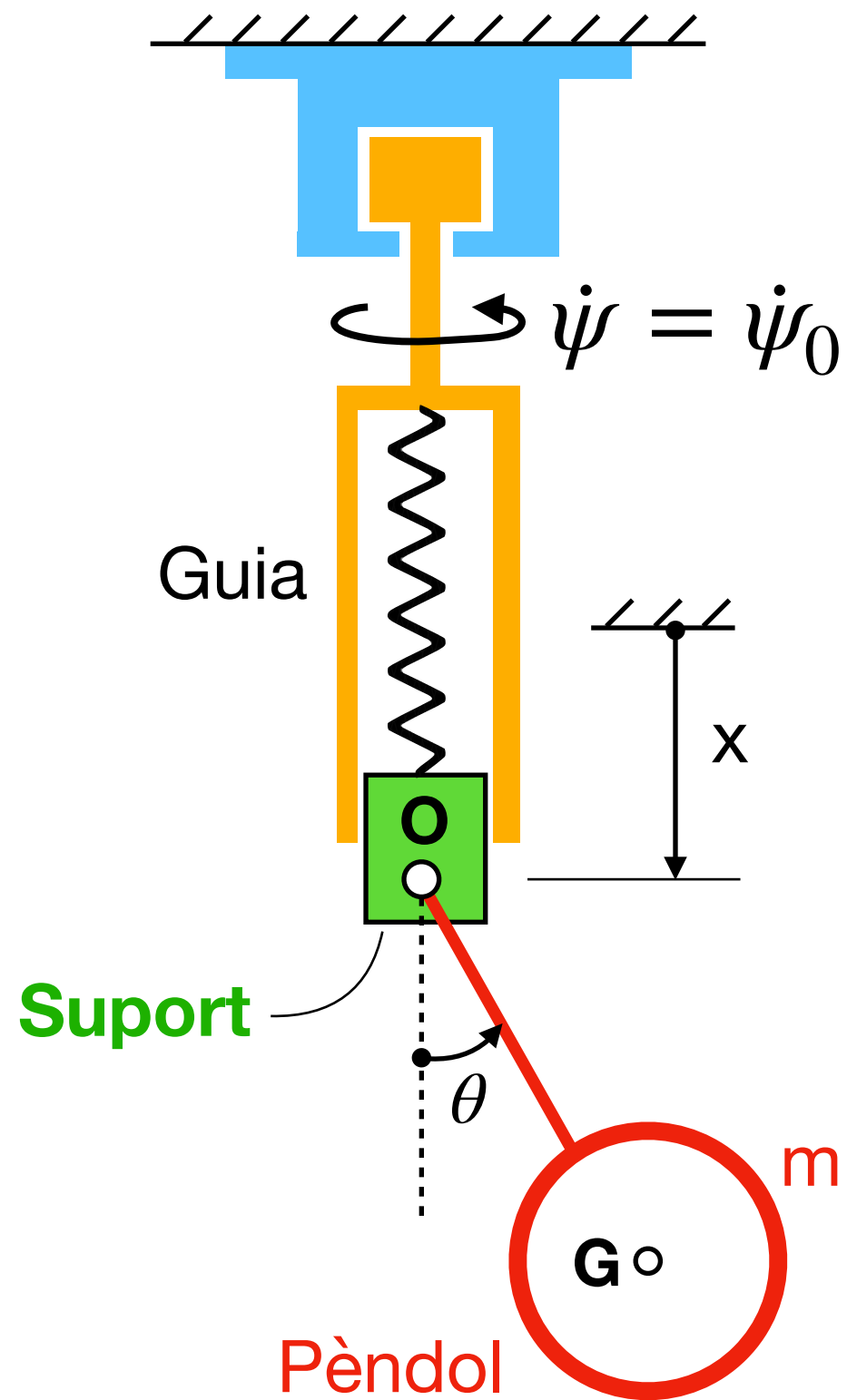


Amb motor aturat: $x = 0$, $\theta = 0$ és config. d'equilibri

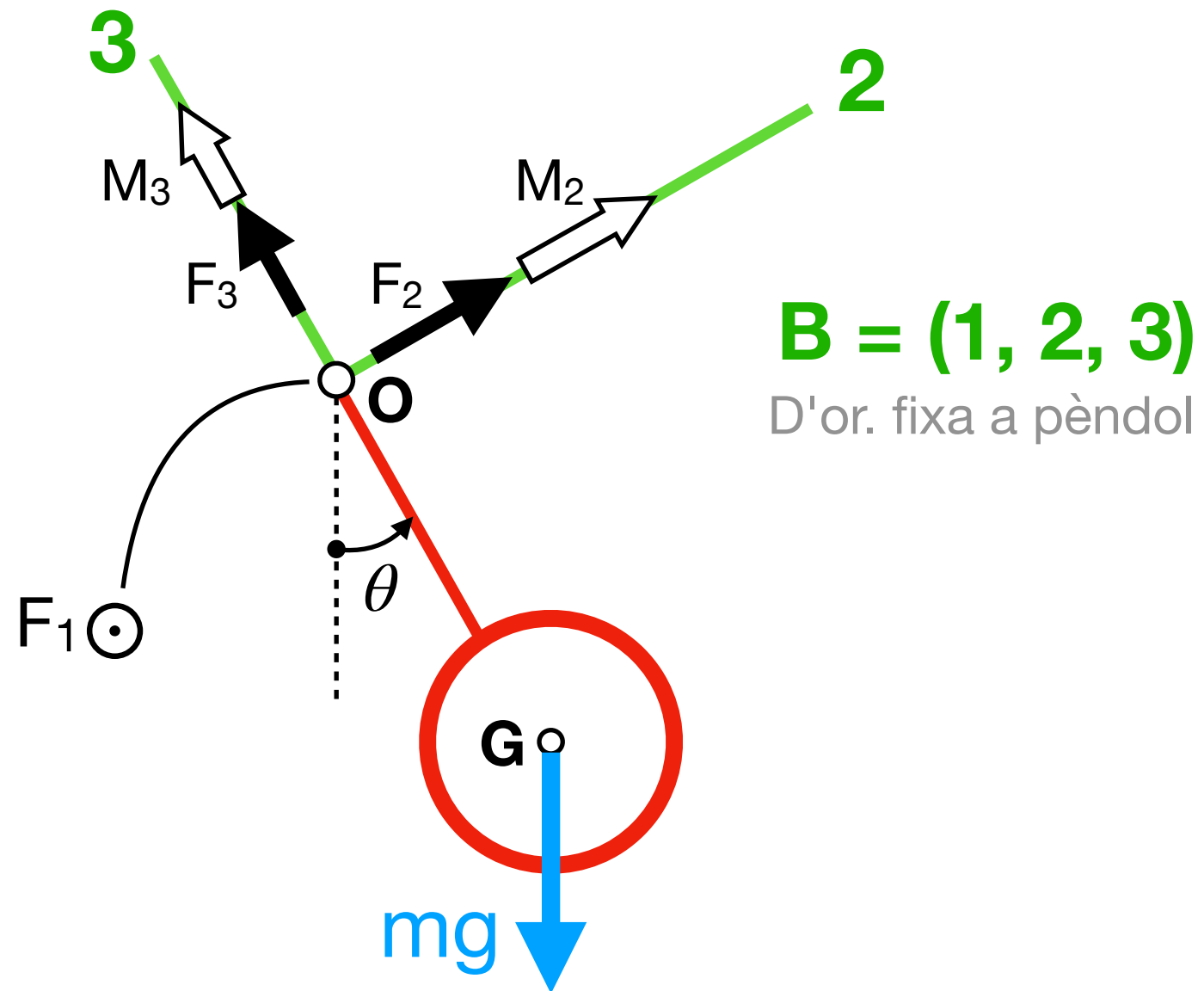
DGI



DGIE



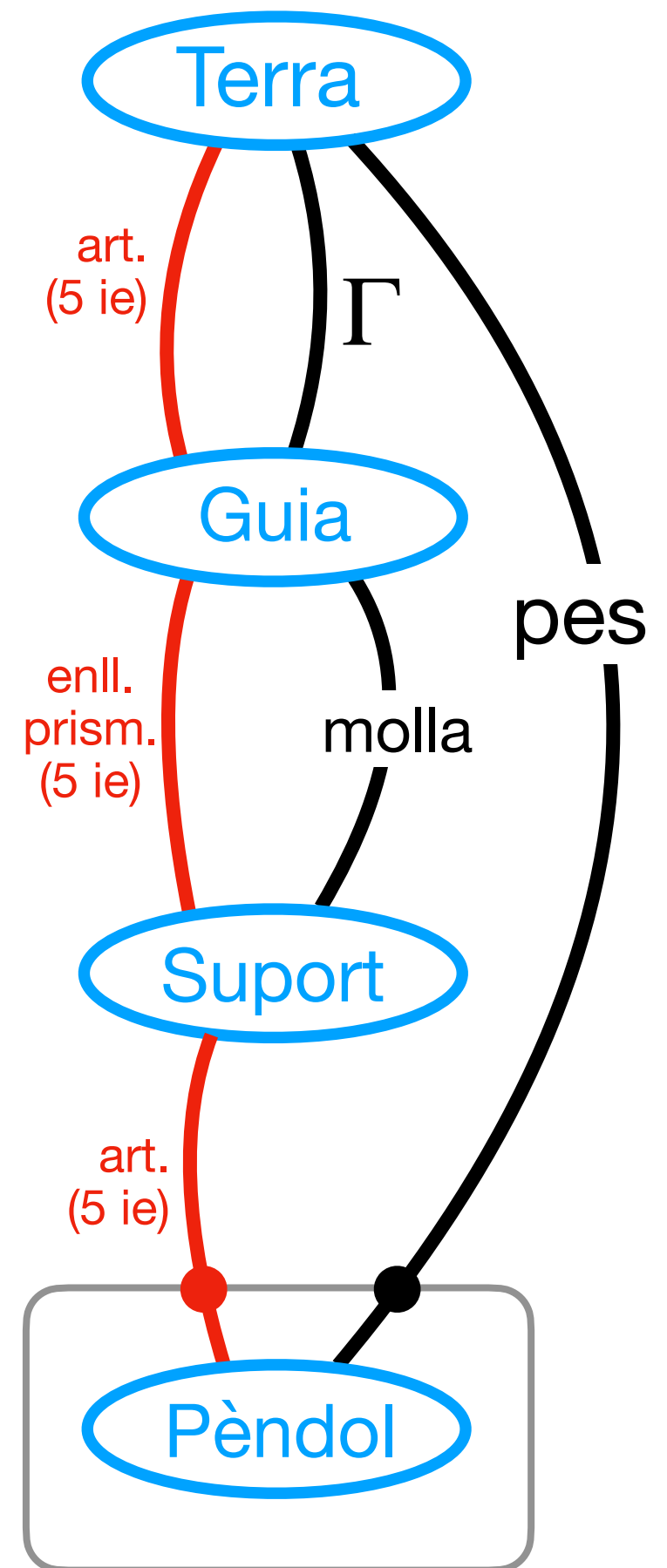
Sist = Pèndol



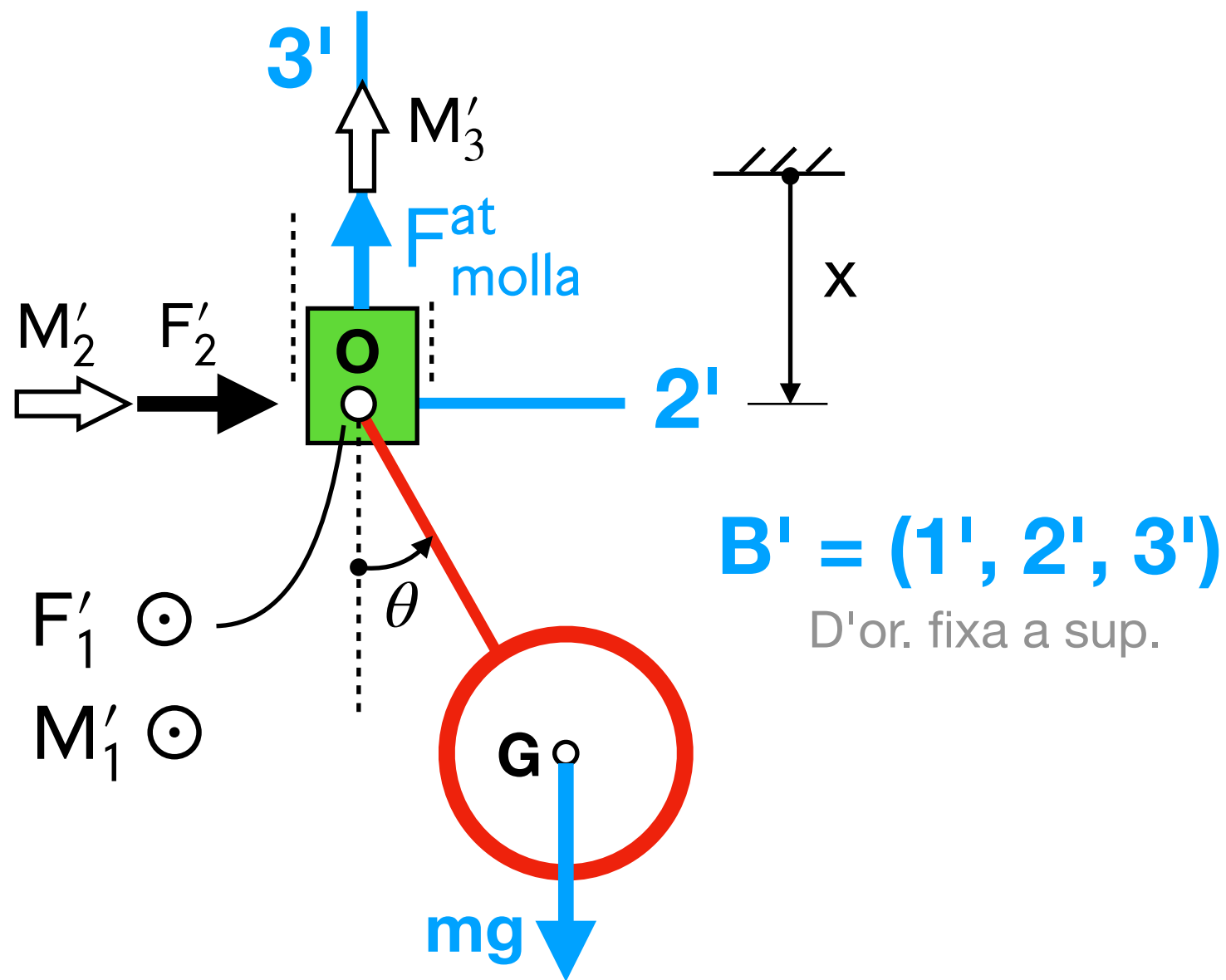
$$\left\{ \bar{F}_{\text{Sup} \rightarrow \text{Pendol}} \right\}_B = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \end{Bmatrix}$$

$$\left\{ \bar{M}_{\text{Sup} \rightarrow \text{Pendol}}(\mathbf{O}) \right\}_B = \begin{Bmatrix} 0 \\ M_2 \\ M_3 \end{Bmatrix} \leftarrow \text{TMC}(\mathbf{O}) \Big|_1$$

lliure de ie



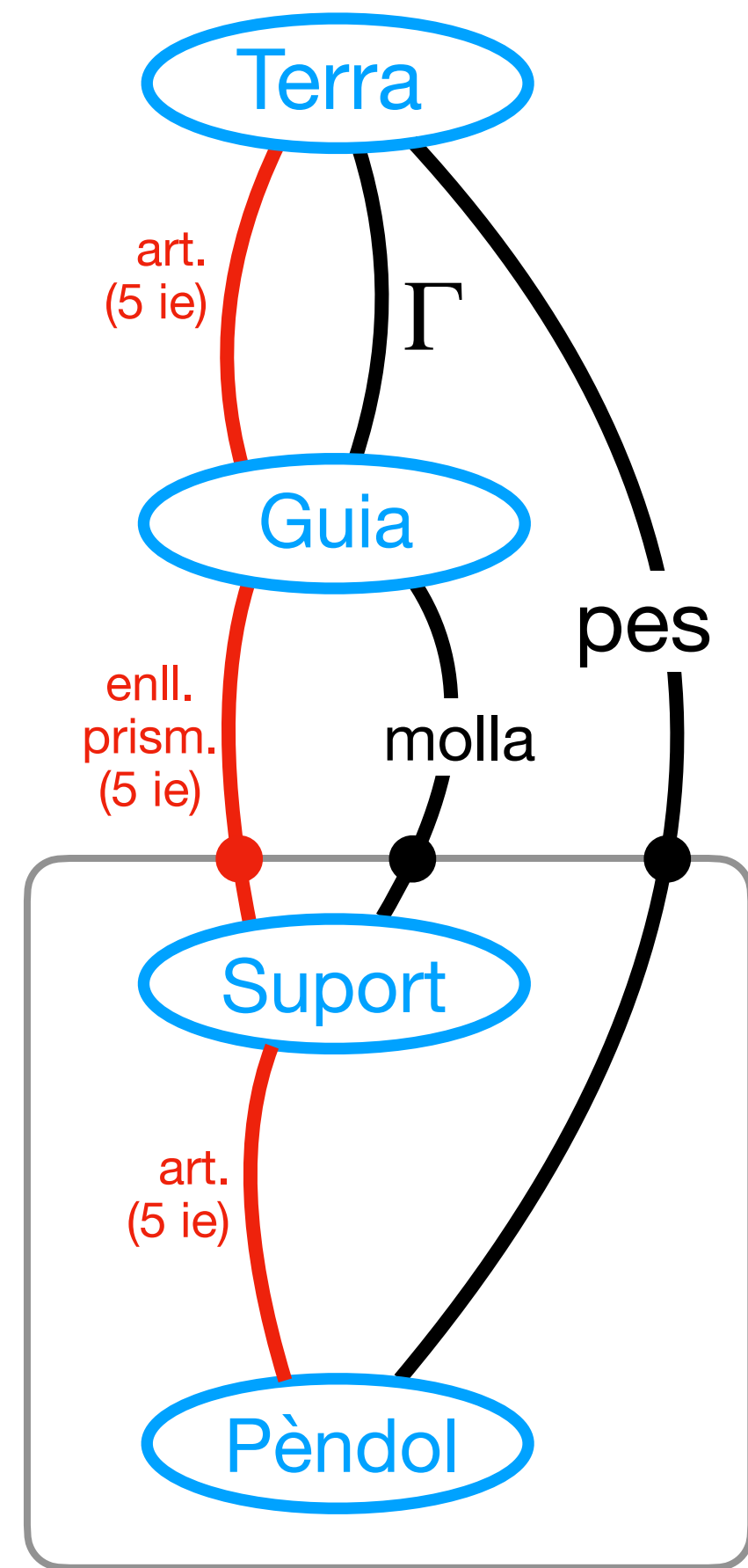
Sist = Pèndol + Suport



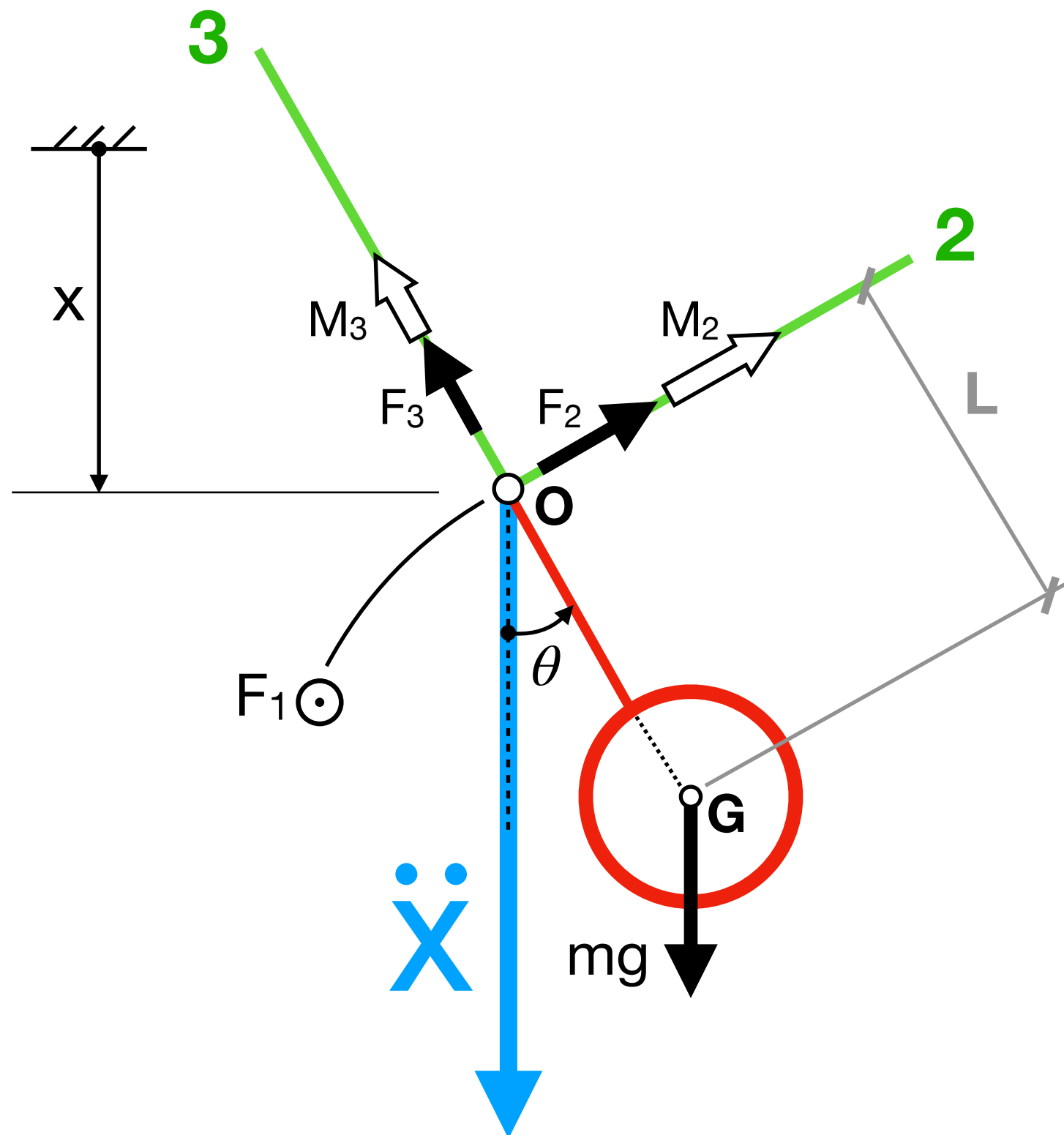
$$\left\{ \bar{F}_{\text{Guia} \rightarrow \text{Sup}} \right\}_{B'} = \begin{Bmatrix} F'_1 \\ F'_2 \\ 0 \end{Bmatrix}$$

$$\left\{ \bar{M}_{\text{Guia} \rightarrow \text{Sup}}(\mathbf{O}) \right\}_{B'} = \begin{Bmatrix} M'_1 \\ M'_2 \\ M'_3 \end{Bmatrix}$$

TQM $_{3'}$
lliure de ie

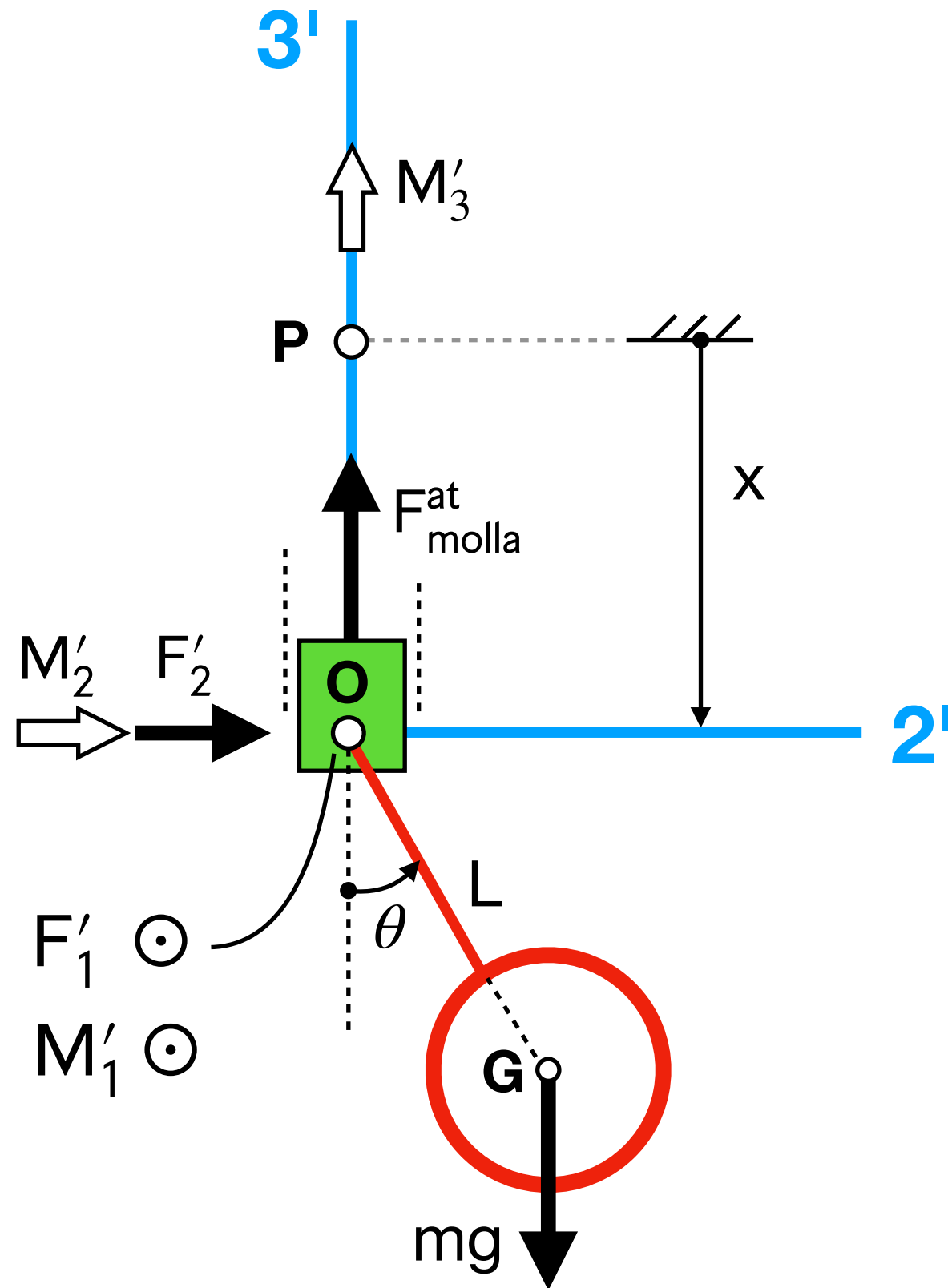


TMC(O)]₁ sobre SIST = Pèndol



TQM]_{3'} per SIST = Pèndol + Suport

$$B' = (1', 2', 3')$$



Pas final per passar a forma explícita

$$\begin{cases} (R^2 + L^2) \ddot{\theta} - (L \sin \theta) \ddot{x} = (L \dot{\psi}_0^2 \cos \theta - g) L \sin \theta \\ - (L \sin \theta) \ddot{\theta} + \ddot{x} = L \dot{\theta}^2 \cos \theta - \frac{k}{m} x \end{cases}$$

Aïllant $\ddot{\theta}$ i $\ddot{x}$

$$\begin{cases} \ddot{\theta} = F_1(\theta, \dot{\theta}, x, \dot{x}) \\ \ddot{x} = F_2(\theta, \dot{\theta}, x, \dot{x}) \end{cases}$$

Eq. mov. θ

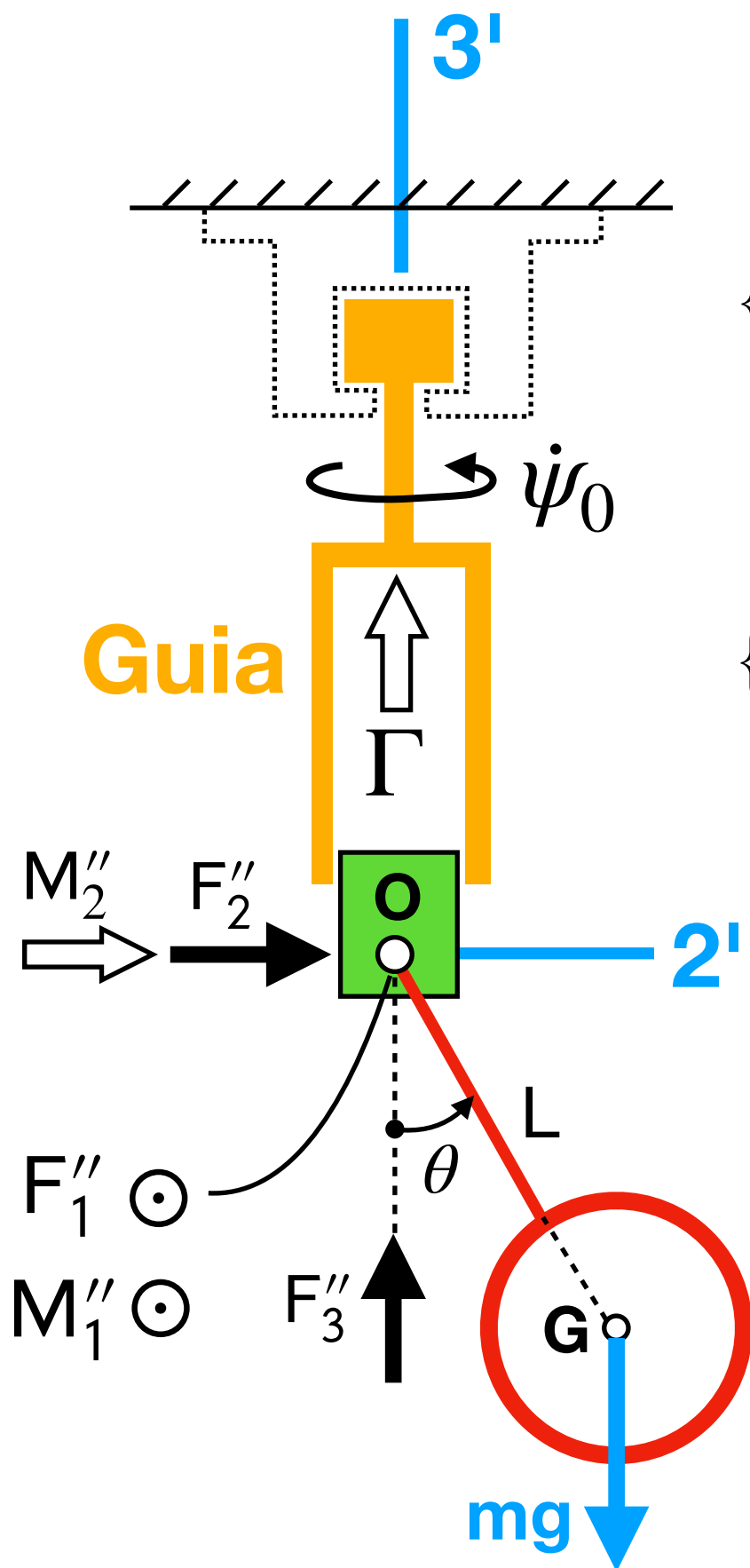
(en forma explícita)

Eq. mov. x

(en forma explícita)

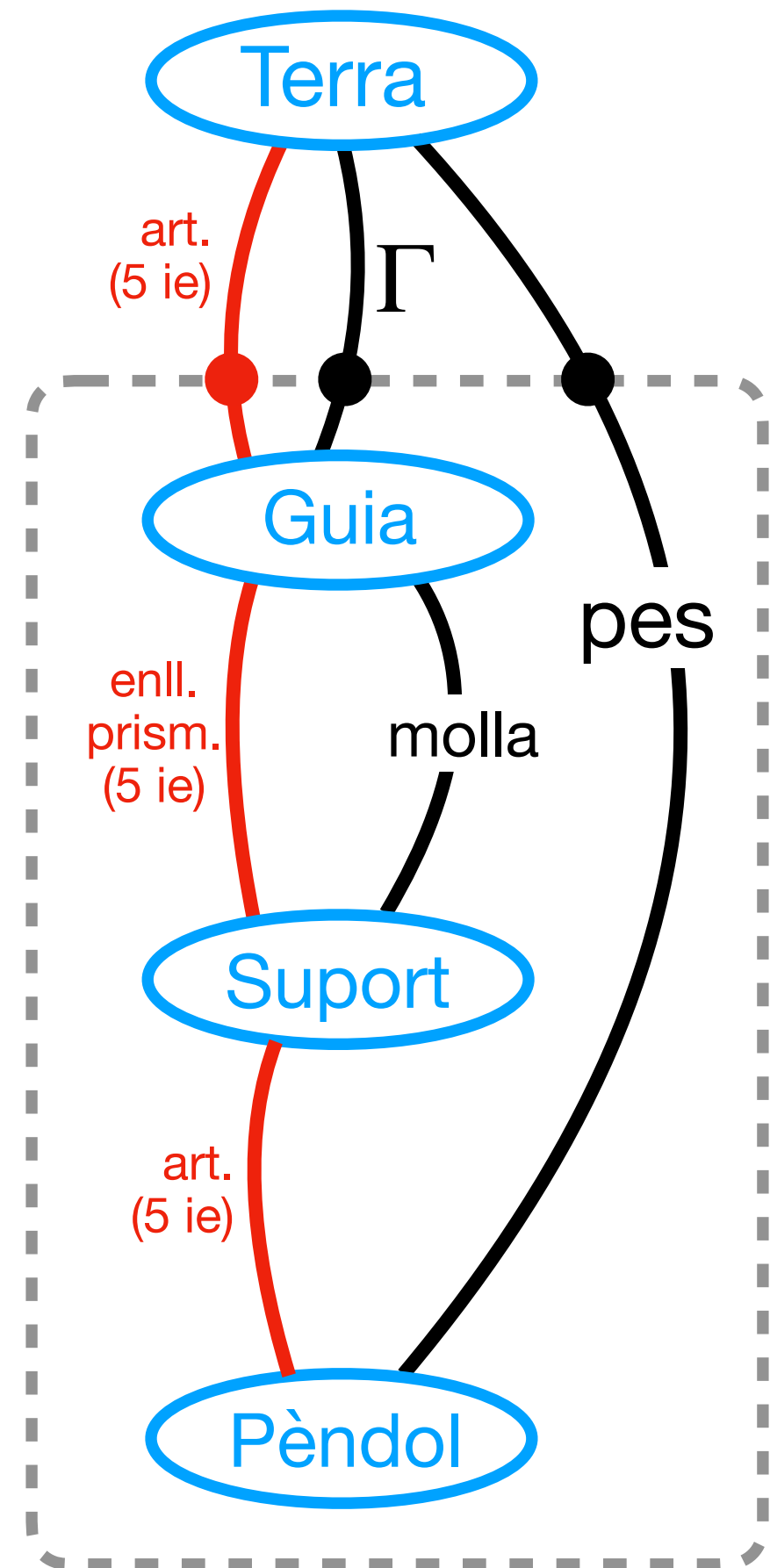
No cal
que el feu

Parell motor per mantenir $\dot{\psi}_0 = ct$



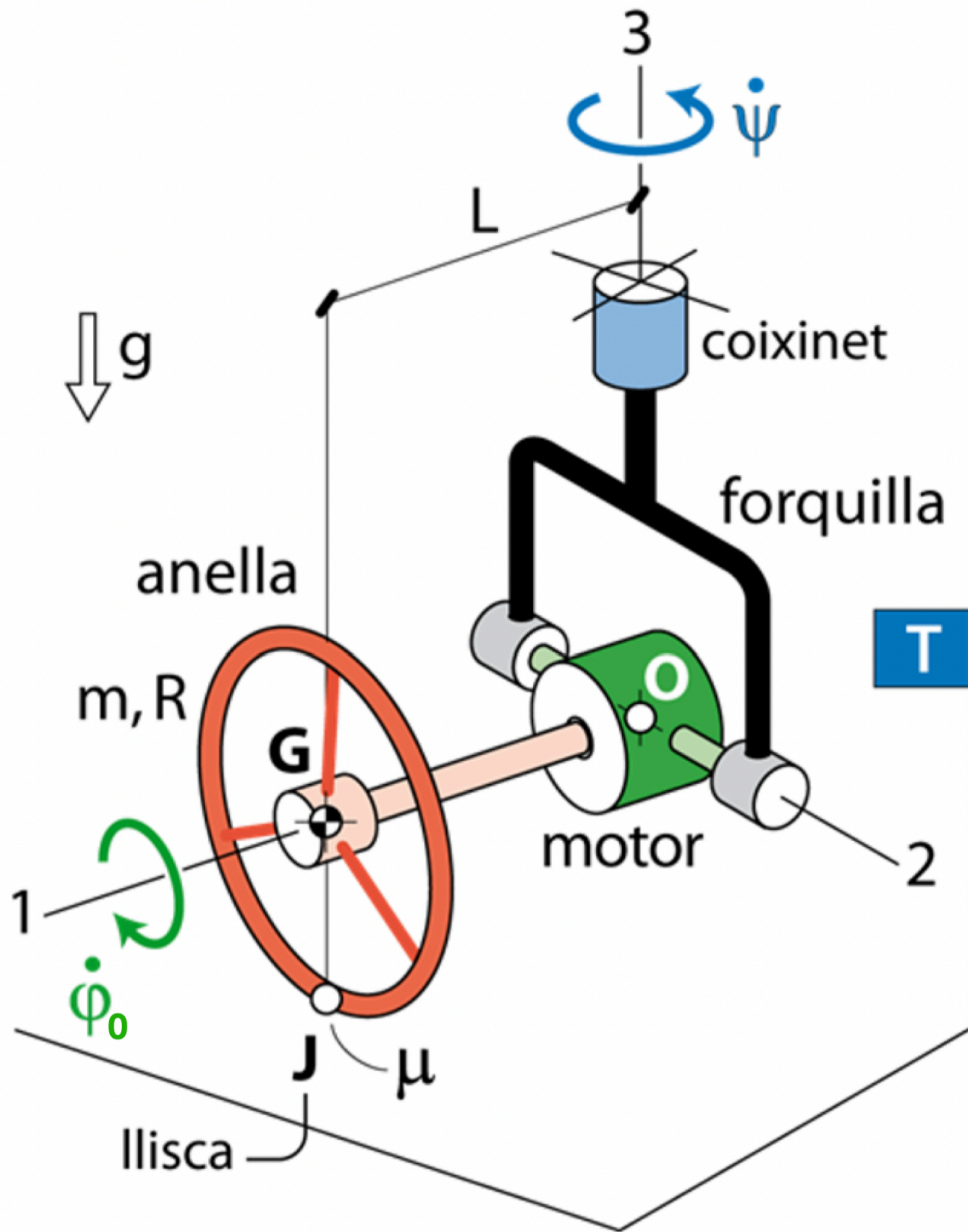
$$\{\bar{\mathbf{F}}_{T \rightarrow \text{Guia}}\}_{B'} = \begin{Bmatrix} F_1'' \\ F_2'' \\ F_3'' \end{Bmatrix}$$

$$\{\bar{\mathbf{M}}_{T \rightarrow \text{Guia}}(\mathbf{O})\}_{B'} = \begin{Bmatrix} M_1'' \\ M_2'' \\ 0 \end{Bmatrix}$$



D'abans

$$\left\{ \dot{\ddot{\mathbf{H}}}_{\text{RTO}}(\mathbf{O}) \right\}_{\text{B}} = \begin{Bmatrix} I_{11}\ddot{\theta} + (I_{33} - I_{22})\dot{\psi}_0^2 \sin\theta \cos\theta \\ (I_{11} + I_{22} - I_{33})\dot{\psi}_0 \dot{\theta} \cos\theta \\ (I_{22} - I_{11} - I_{33})\dot{\psi}_0 \dot{\theta} \sin\theta \end{Bmatrix} = \begin{Bmatrix} \dot{\mathbf{H}}_1 \\ \dot{\mathbf{H}}_2 \\ \dot{\mathbf{H}}_3 \end{Bmatrix}$$

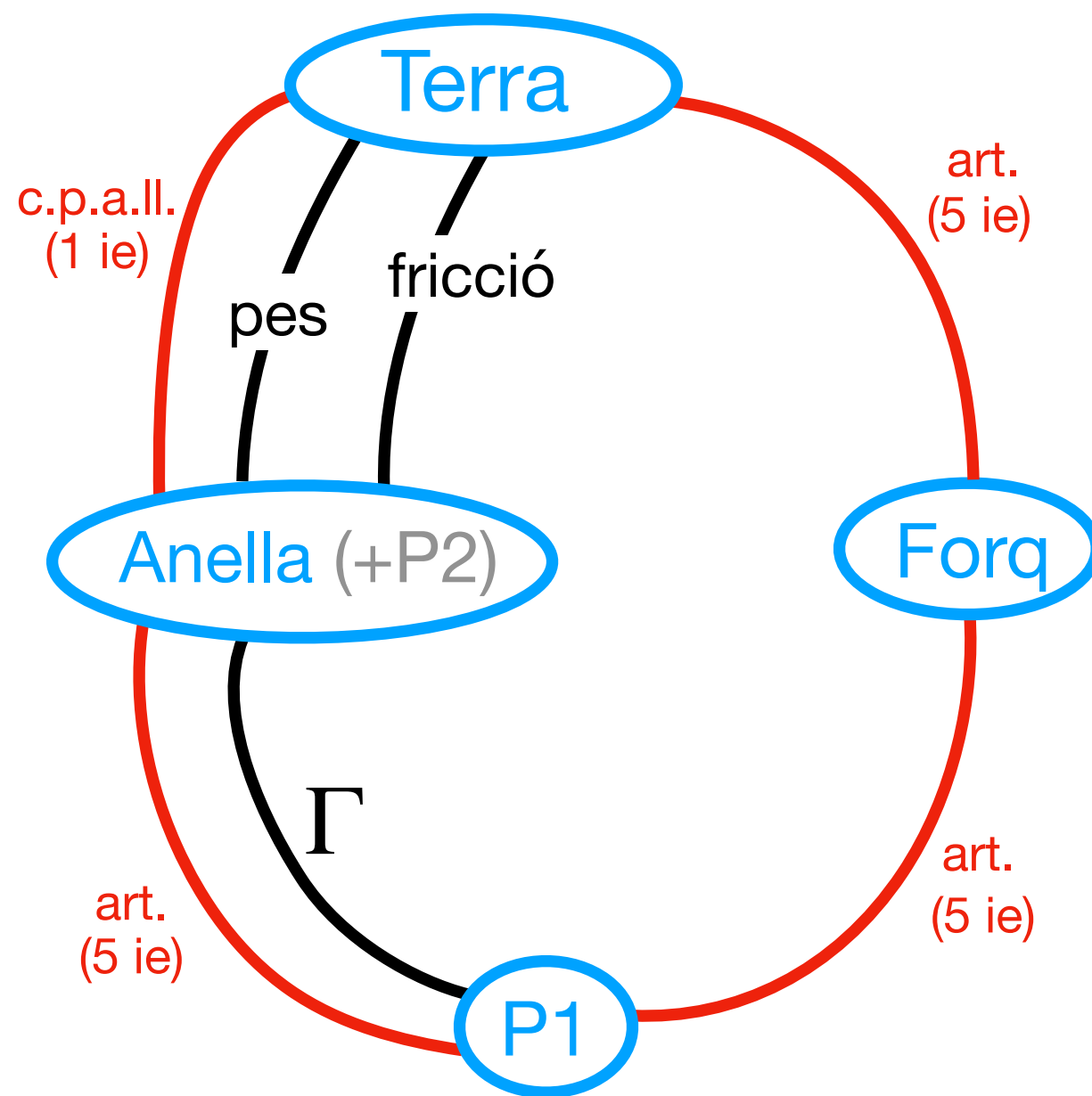
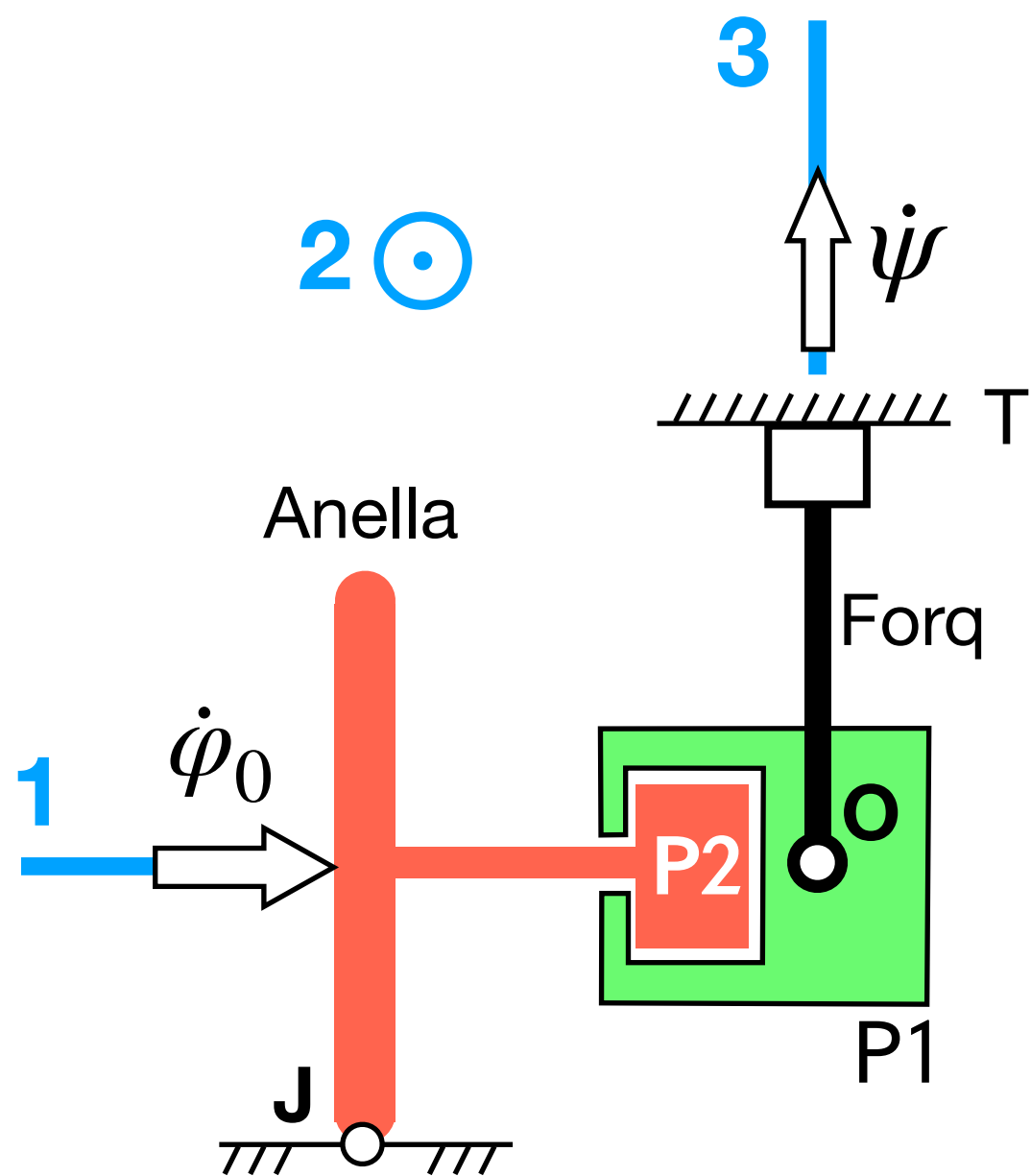


- GL sistema?
- DGI
- Caracterització de torsors?
- Eq. mov per a la coord. ψ
- Força normal a J
- Parell motor per mantenir $\dot{\phi}_0$

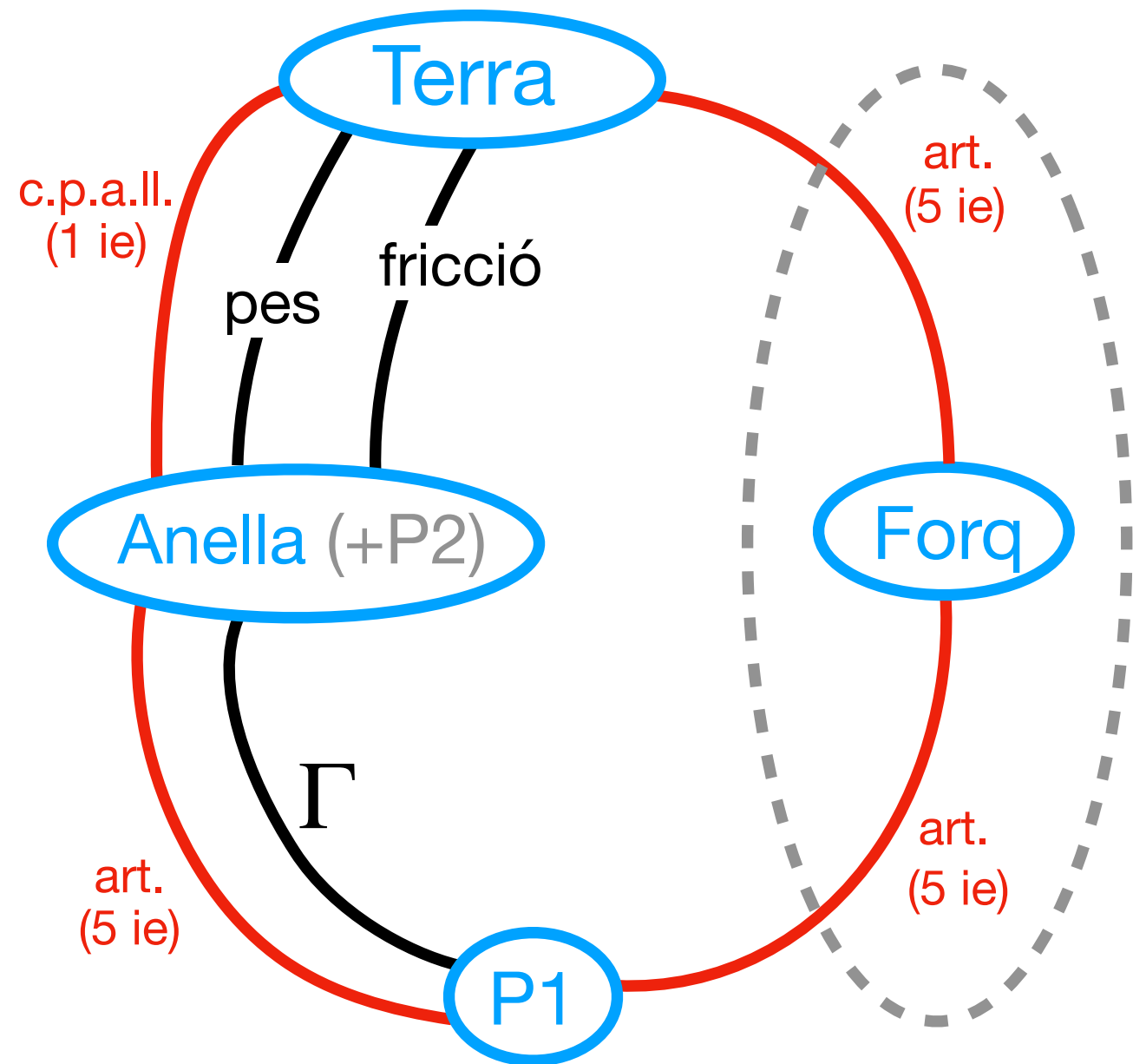
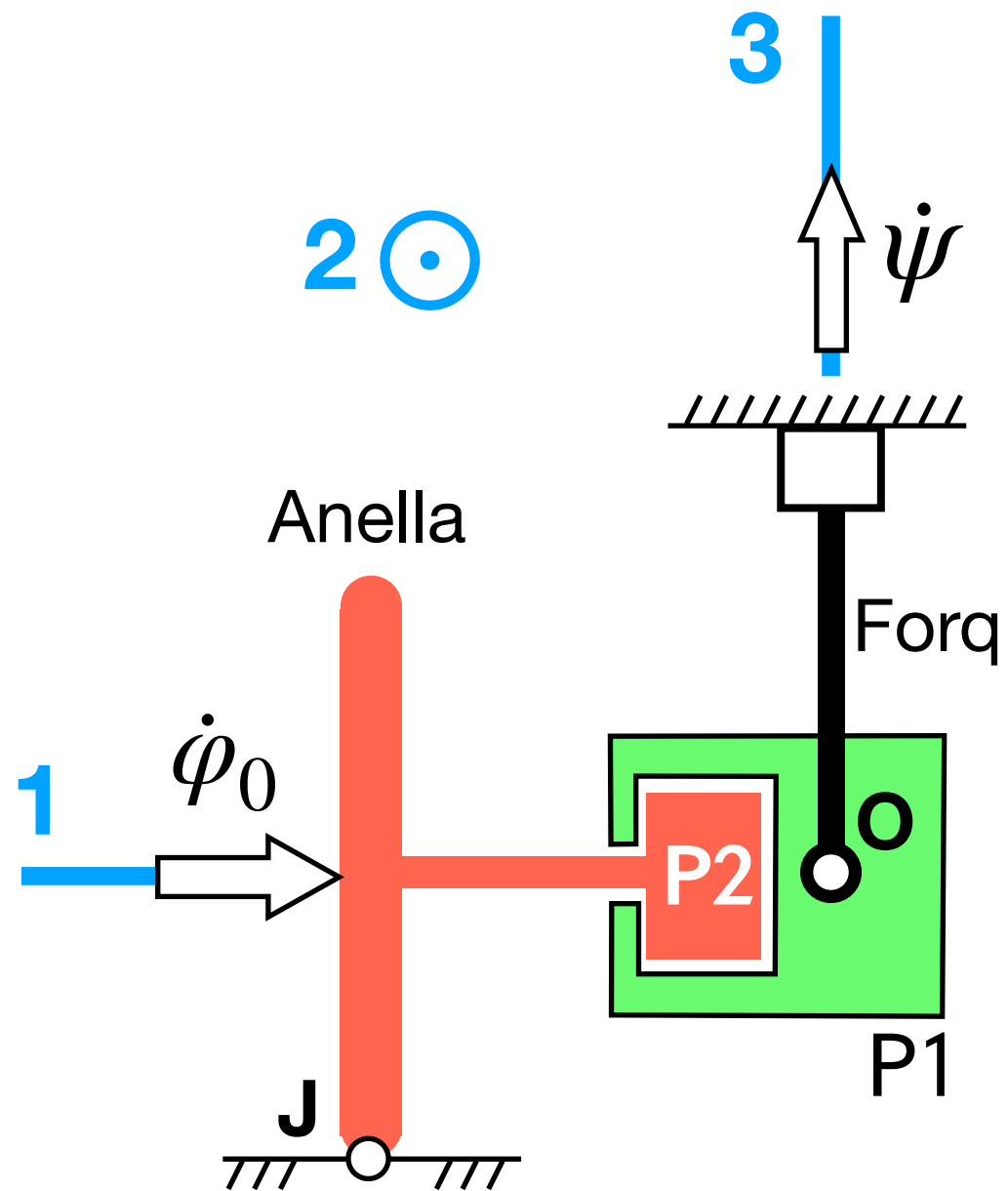
Manté contacte puntual a J

Motor manté $\dot{\phi}_0 = \text{ct}$

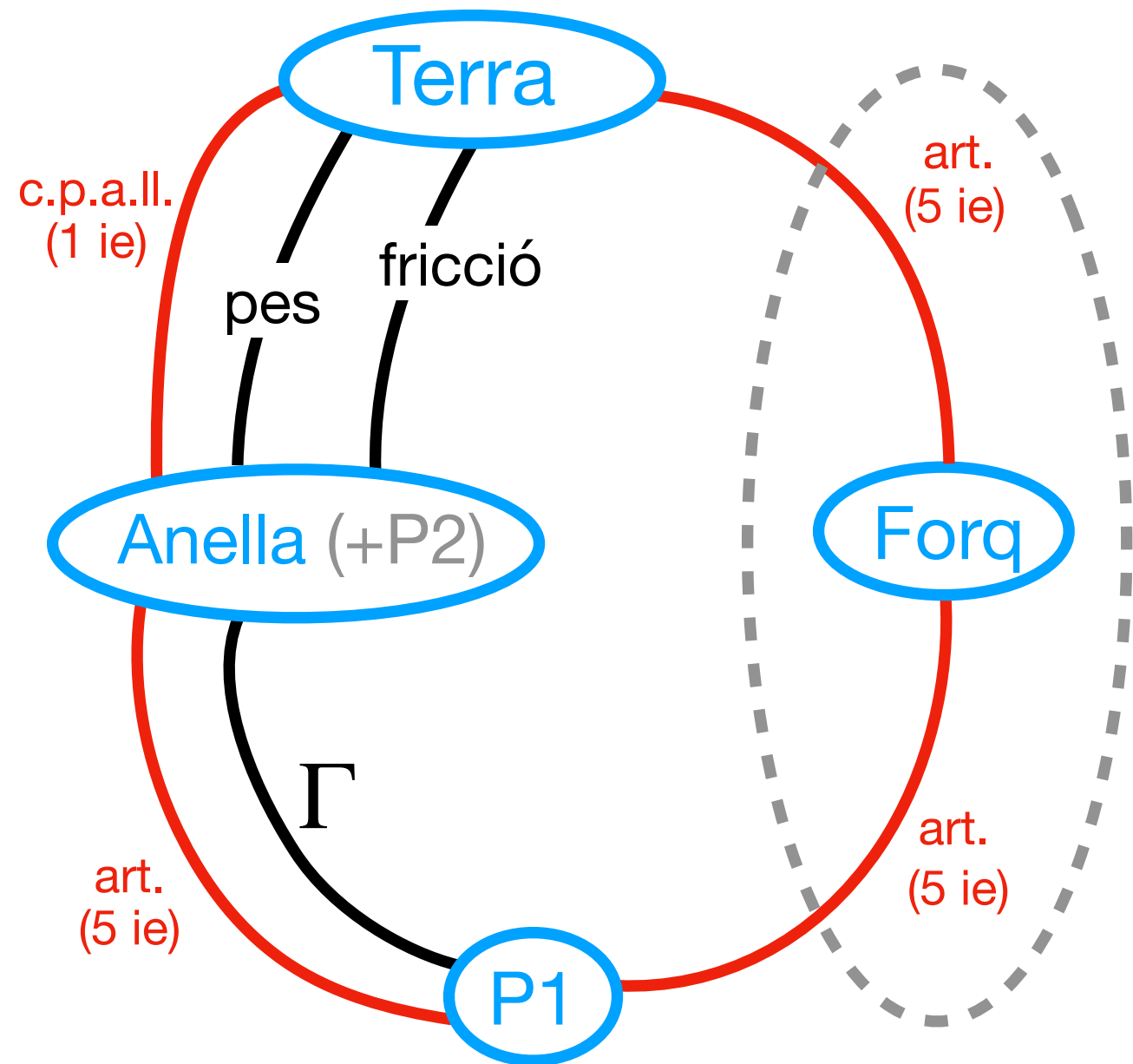
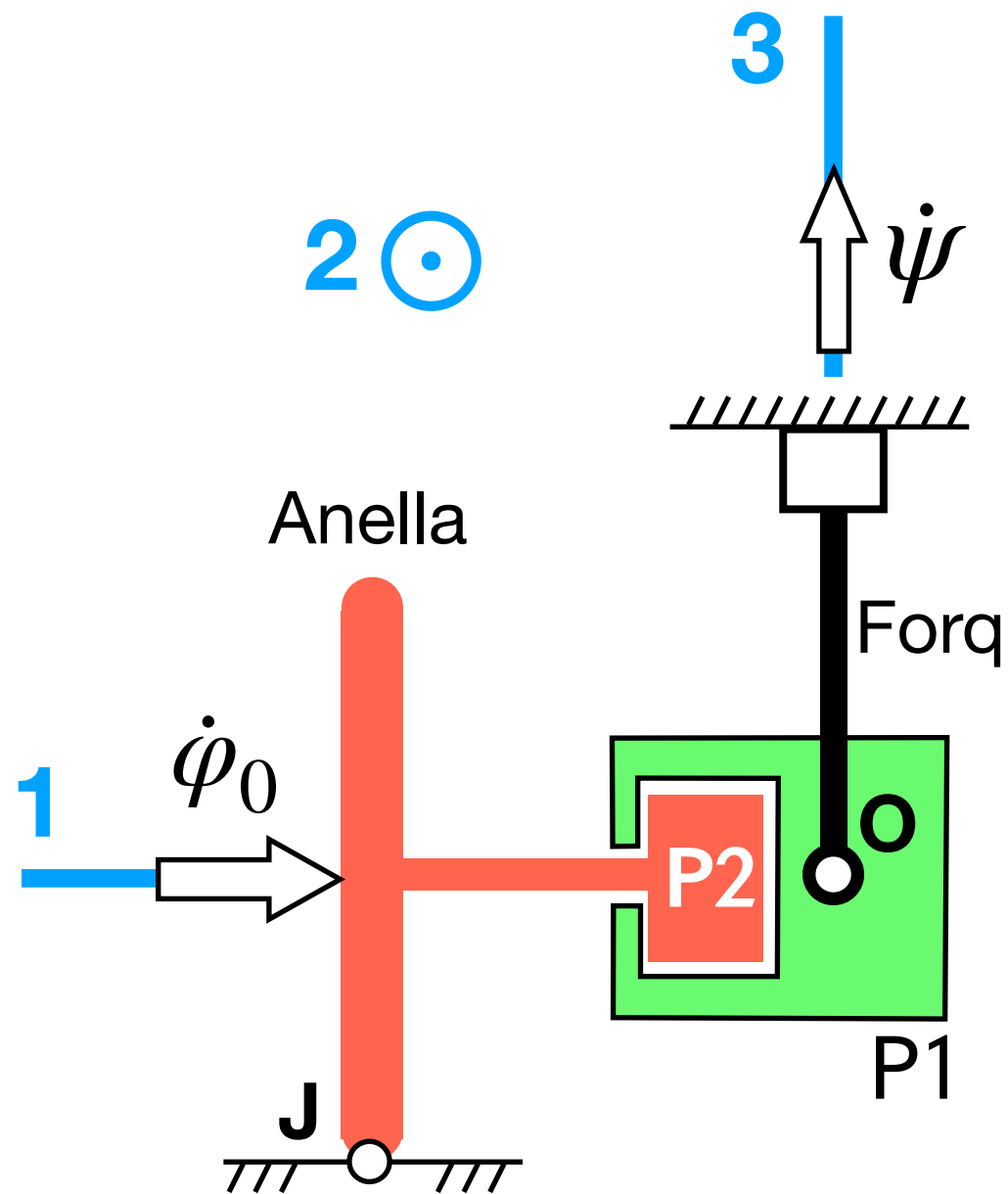
Inicialment: $\psi = 0, \dot{\psi} = 0$



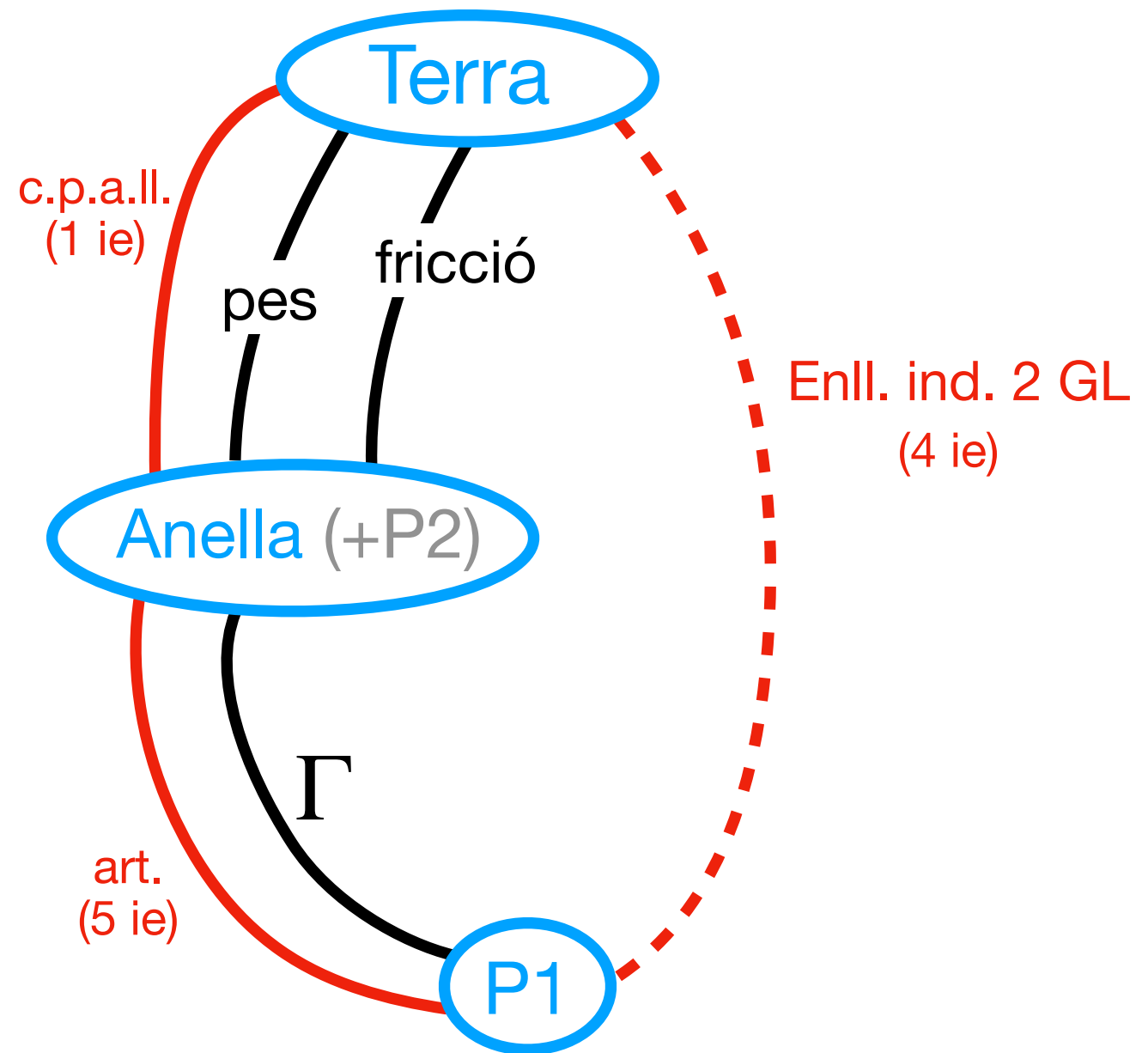
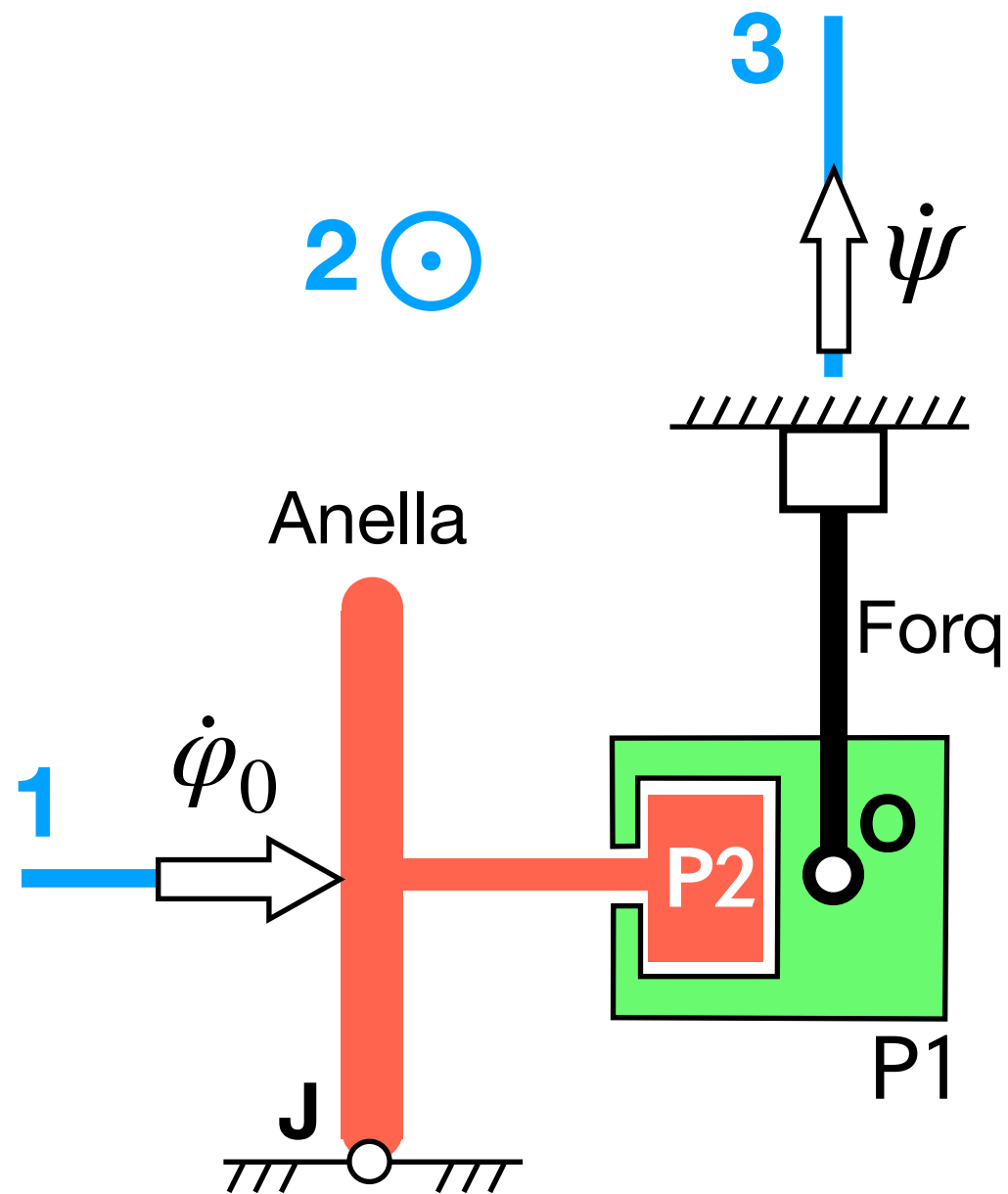
Forq és SAE ...



Substituim **Forq** per enllaç indirecte $T \rightarrow P_1$



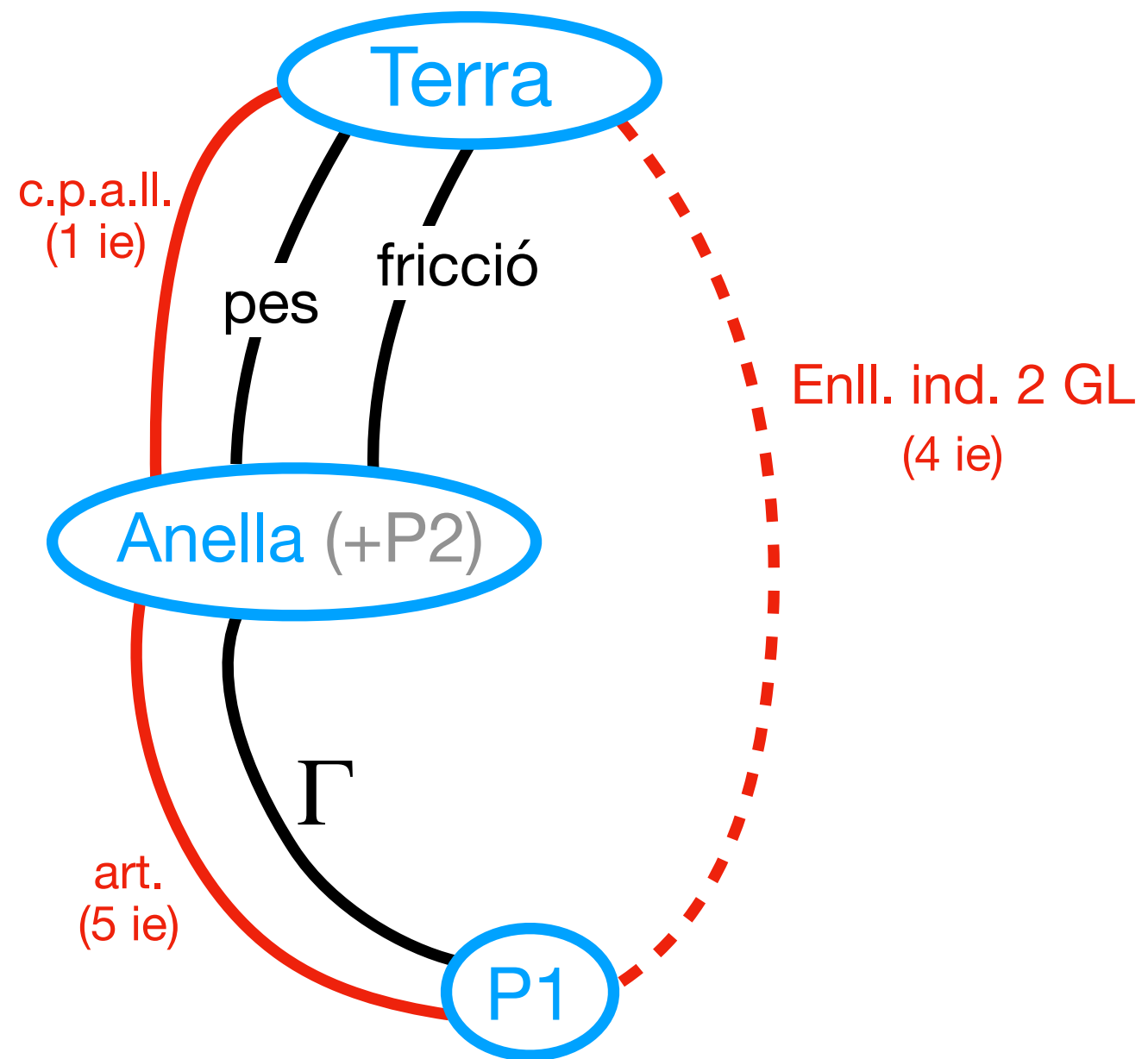
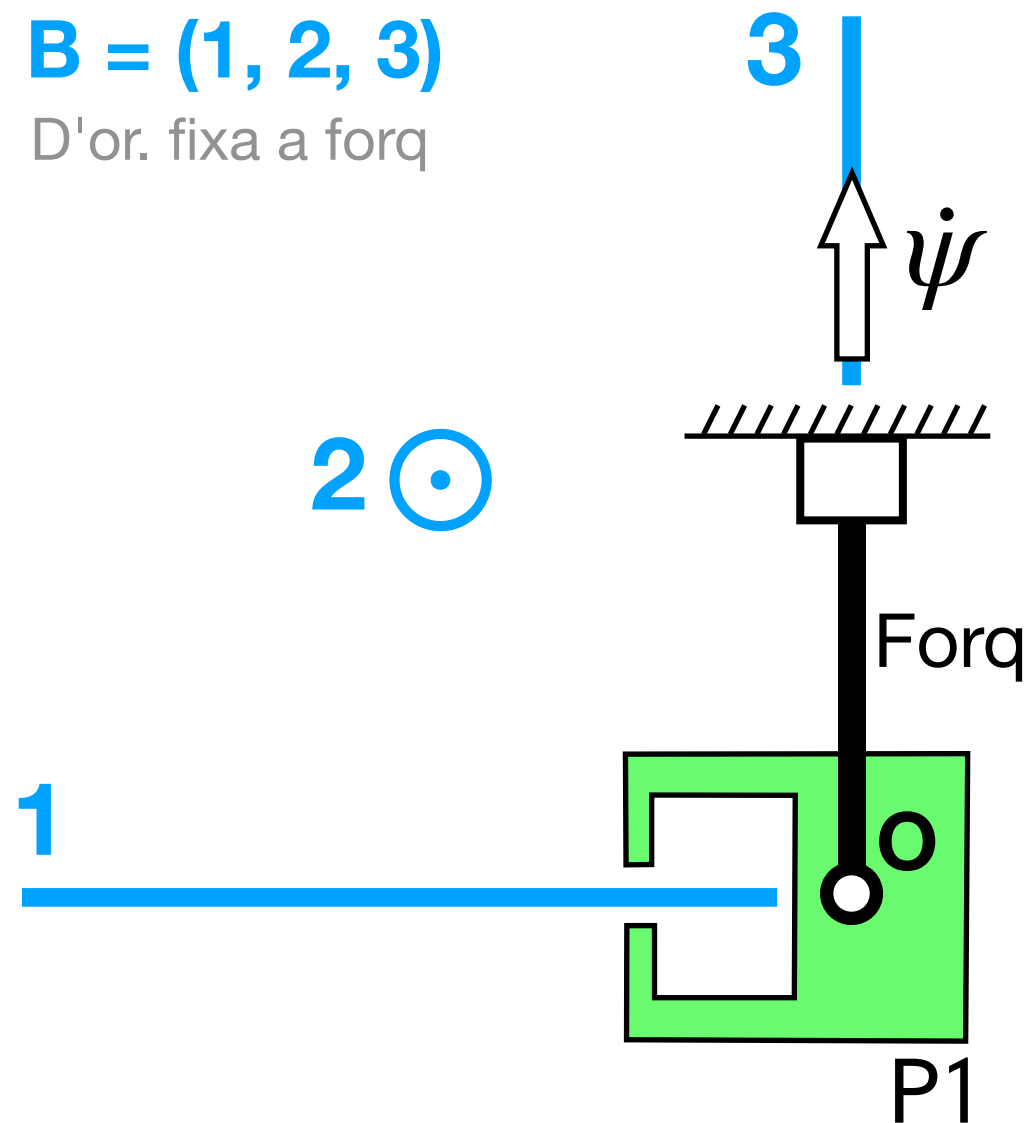
Substituim **Forq** per enllaç indirecte $T \rightarrow P_1$



Substituïm **Forq** per enllaç indirecte $T \rightarrow P_1$

B = (1, 2, 3)

D'or. fixa a forq

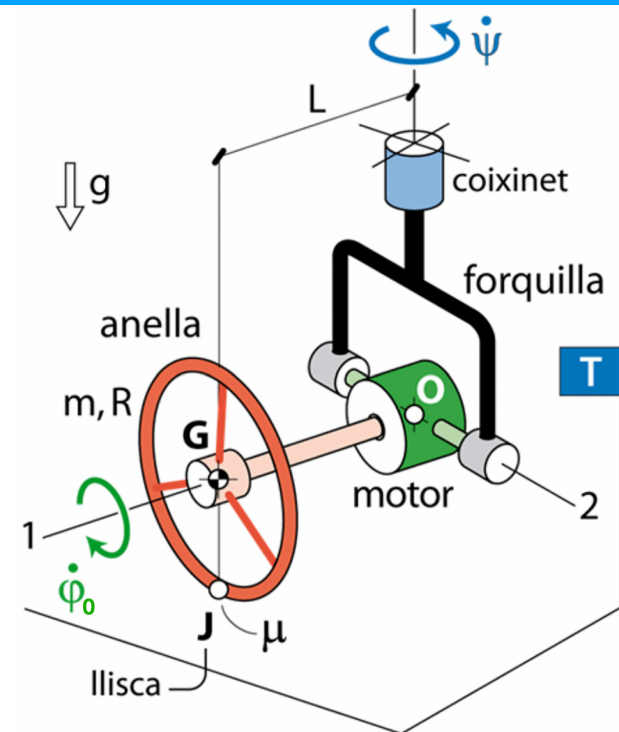


Torsor d'enllaç indirecte $T \rightarrow P1$ al punt **O**:

$$\left\{ \bar{\mathbf{F}}_{T \rightarrow (\text{forq}) \rightarrow P1} \right\}_B = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \end{Bmatrix} \quad \mathbf{O} \text{ fix a } T$$

$$\left\{ \bar{\mathbf{M}}_{T \rightarrow (\text{forq}) \rightarrow P1} (\mathbf{O}) \right\}_B = \begin{Bmatrix} M_1 \\ 0 \\ 0 \end{Bmatrix} \quad \begin{array}{l} 0 \text{ en les rotacions} \\ \text{permeses de } P1 \\ \text{resp } T \end{array}$$

Full ruta per eq. del mov. ψ

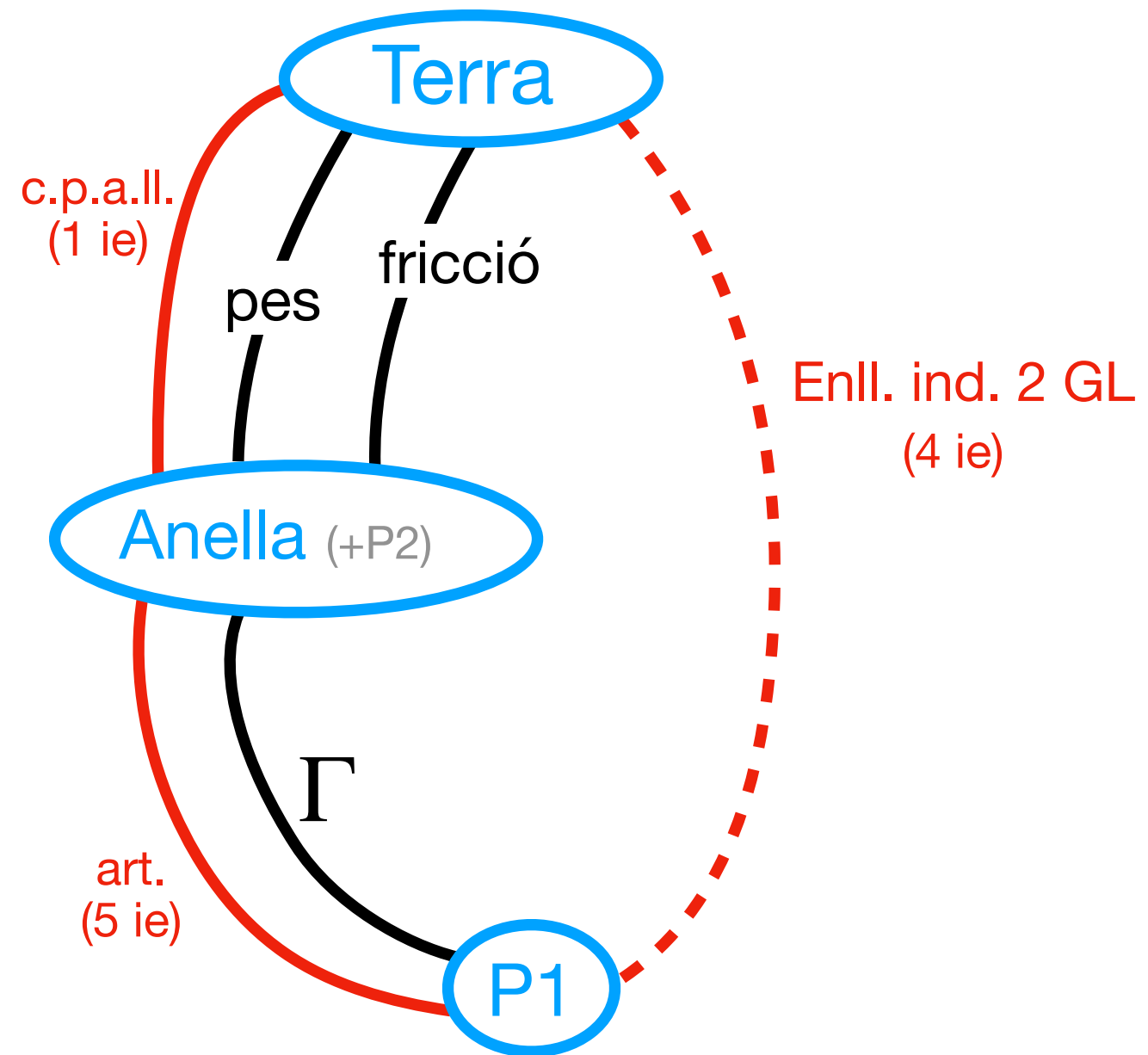


ψ afecta la cinemàtica de **P1** i **Anella**



Sist ha d'incloure **P1** o **Anella** (o ambdós)

Sistema	Incògn.	Problema
Anella	6 ie, Γ , $\ddot{\psi}$	INDET
P1	9 ie, Γ , $\ddot{\psi}$	INDET
Anella + P1	5 ie, $\ddot{\psi}$	DET



$\Rightarrow$ Triem SIST = Anella + P1